

NAMMA CLINIC

Digital Healthcare & Clinic Management Platform

Comprehensive Client Application Reference Manual

Target Audience: DHO, Health Dept Management, Administrators, Doctors, Nurses, Front Desk, Pharmacists, Lab Techs	Operational Scope: Primary Healthcare Centers (PHCs), Urban Health Centers, Namma Clinics
Deployment Mode: Local Clinic Workstations & Laptops (Zero Cloud Dependency)	Document Version: Release 2026.1 (Phase DOC-02 Client Reference)

PART A - UNDERSTANDING THE APPLICATION

1. Document Overview & Executive Introduction

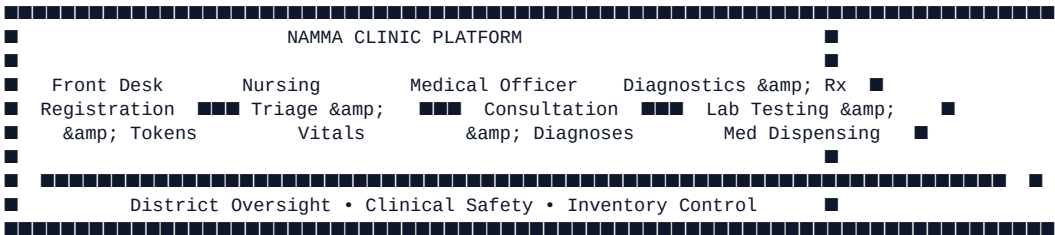
This document provides a comprehensive, non-technical reference guide to the **Namma Clinic Platform**. It is written for healthcare administrators, medical superintendents, clinical leaders, and frontline healthcare professionals.

Whether you are a District Health Officer reviewing network performance, a Medical Officer conducting consultations, a Staff Nurse prioritizing walk-in patients, a Pharmacist dispensing medications, or a Front Desk Officer welcoming citizens at reception, this guide explains how the platform supports your daily work and how patients move safely and efficiently through primary care.

2. What is Namma Clinic?

Namma Clinic is a unified digital health and clinic operations platform created specifically for primary healthcare facilities, including urban health centers, neighborhood clinics, and primary health centers (PHCs).

The platform connects all healthcare disciplines within a clinic - reception, nursing, clinical examination, laboratory diagnostics, and pharmacy dispensing - into one synchronized workflow. Every action taken by a staff member immediately updates the live queue, alerts relevant colleagues, and creates an audit-ready medical record.



The system is designed to run directly on standard clinic workstations and laptops at the clinic facility itself. It does not require continuous high-speed internet to perform routine clinic tasks, guaranteeing that citizen care continues without interruption even during local internet disruptions.

3. Problems Namma Clinic Solves

Urban primary clinics face significant operational hurdles when relying on manual registers, paper prescription slips, and fragmented single-purpose applications:

Traditional Clinic Challenge	Namma Clinic Operational Solution
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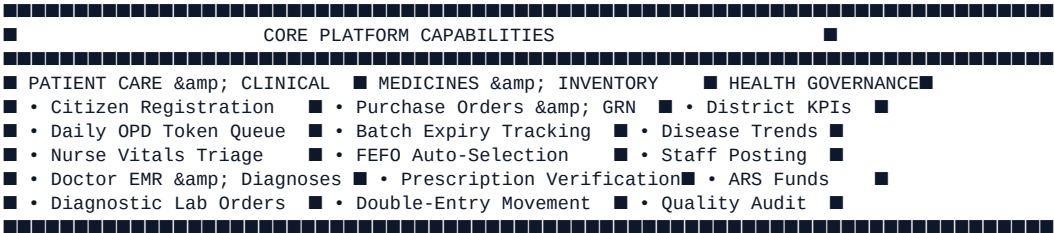
Overcrowded Waiting Rooms & Queue Disputes Patients congregate around consultation doors without knowing when they will be seen.	Automated Sequential OPD Tokens Every patient receives an individual daily token upon arrival. Workstations call patients sequentially, providing fair and orderly patient flow.
Unrecognized Emergencies in Waiting Areas Critically ill walk-in patients wait alongside non-urgent cases without prompt clinical review.	Standardized Nurse Triage & Acuity Flags Nurses measure vitals before the doctor consultation. Critical cases are flagged red/orange and prioritized in the doctor's queue.
Illegible Hand-Written Prescriptions Pharmacy errors, misread drug dosages, and dispensing mistakes caused by poor handwriting.	Structured Electronic Prescriptions Doctors prescribe from standardized drug lists with verified strengths, dosage forms, frequencies, and duration instructions.
Expired Drug Wastage & Unpredictable Stockouts Expensive medicine batches expire unseen in storage while clinics unexpectedly run out of vital medications.	Automated First-Expired, First-Out (FEFO) Dispensing The system recommends the earliest-expiring active batch first and triggers automated alerts when minimum stock thresholds are crossed.
Compromised Patient Medical Privacy Clerical staff and intake workers inadvertently view sensitive clinical notes and diagnoses in paper registers.	Strict Separation of Duties & Privacy Boundaries Front-desk personnel only view demographic records. Clinical consultation history, diagnoses, and lab results are restricted to doctors and nurses.
Delayed Disease Outbreak Detection District health authorities receive paper disease reports days or weeks after an outbreak has begun.	Real-Time Syndromic Surveillance Every diagnosed fever, respiratory infection, or waterborne illness is instantly plotted on district outbreak dashboards.

4. Who Uses Namma Clinic?

The platform provides dedicated, custom-tailored workstations for eight operational roles:

1. **Front Desk Officer:** Welcomes citizens, searches existing patient records, registers new patients, and issues sequential daily outpatient (OPD) tokens.
2. **Staff Nurse:** Measures vital signs, computes Body Mass Index (BMI), assigns triage acuity scores, records follow-up care, and coordinates community health camps.
3. **Medical Officer (Doctor):** Examines patients, documents clinical notes and diagnoses, requisitions diagnostic laboratory tests, writes electronic prescriptions, and arranges specialist referrals.
4. **Diagnostic Lab Technician:** Receives doctor-ordered laboratory tests, collects patient specimens, inputs test values, and publishes verified laboratory investigation reports.
5. **Pharmacist:** Verifies doctor prescriptions against available facility stock, dispenses medications according to First-Expired, First-Out rules, and counsels patients on drug usage.
6. **Inventory Officer:** Manages the clinic medicine store, generates purchase orders to suppliers, inspects incoming deliveries with Goods Receipt Notes (GRN), and audits stock balances.
7. **Clinic Administrator:** Manages facility operational schedules, configures clinic departments, onboards staff, assigns operational roles, and reviews daily attendance reports.
8. **District Health Officer (DHO):** Oversees public health across all networked clinics in the district, monitors disease clusters, reviews clinic quality ratings, and governs staff allocations.

5. Major Platform Capabilities



- **Universal Master Patient Index:** Rapid search by mobile number, patient name, or national health ID prevents duplicate paper files and maintains a single longitudinal medical record.

- **Five-Tier Clinical Acuity Triage:** Standardized prioritization (IMMEDIATE, VERY_URGENT, URGENT, STANDARD, NON_URGENT) ensures acute conditions receive rapid care.
- **Closed-Loop Diagnostic Flow:** Test orders created in the consultation room appear instantly on the lab technician's screen; completed results flow directly back to the doctor.
- **Warehouse vs. Dispensary Separation:** Warehouse storekeepers cannot dispense medicines to patients; dispensary pharmacists cannot arbitrarily modify bulk warehouse balances without verified receiving documentation.
- **Full Historical Audit Trails:** Every clinical diagnosis, prescription item, token cancellation, and stock deduction is stamped with the author's name, time, and facility location.

6. How the Overall Clinic Operates

In a typical clinic morning, all healthcare disciplines work in harmony through connected platform screens:

1. **8:00 AM - Storekeeper & Pharmacist Check:** The Inventory Officer and Pharmacist check the alerts screen to review near-expiry batches and verify stock levels for the day.
2. **8:30 AM - Front Desk Intake Starts:** As citizens arrive, the Front Desk Officer looks up existing cards or registers new patients, immediately issuing daily token numbers.
3. **8:35 AM - Triage Commences:** The Staff Nurse calls waiting tokens into the triage room, checks blood pressure, temperature, and pulse, and marks the clinical acuity.
4. **9:00 AM - Doctor Consultations:** The Medical Officer calls triaged patients in order of urgency, reviews vitals, conducts clinical examinations, and enters treatment plans.
5. **9:20 AM - Investigations & Dispensing:** Patients requiring tests proceed to the laboratory bench. Patients with prescriptions collect their medications at the pharmacy counter.
6. **1:30 PM - Afternoon Review:** The Clinic Administrator and District Health Officer review real-time clinic throughput, waiting times, and disease surveillance summaries.

PART B - USER WORKSTATIONS

1. District Health Officer (DHO) Workstation

Purpose

The **District Health Officer Console** provides comprehensive executive governance, disease surveillance, and administrative supervision over all Primary Health Centers and Urban Clinics in the district.

Core Responsibilities

- Monitor district-wide patient turnout, clinic operational status, and waiting times.
- Track real-time communicable disease trends and syndromic outbreak alerts.
- Supervise clinical quality scores, patient satisfaction metrics, and guideline adherence.
- Oversee staff deployment, postings, and transfers between district facilities.
- Supervise Arogya Raksha Samiti (ARS) development fund balances and expenditures.

Screens Available

1. **District Dashboard:** High-level network KPIs, facility occupancy, and active caseload.
2. **Healthcare Network Map:** Geographic representation of clinics, health centers, and referral hospitals.
3. **Public Health Surveillance:** Disease incidence curves, syndromic fever tracking, and cluster warnings.
4. **District Staff Directory:** Comprehensive personnel roster across all district health facilities.
5. **Quality & Compliance Hub:** Clinic evaluation indicators, inspection checklists, and statutory licenses.

Step-by-Step Workflow

1. The DHO logs in and reviews the **District Dashboard** to assess total patient visits across all clinics today.
2. The DHO opens **Public Health Surveillance** to check for unexpected clusters of respiratory illness or acute diarrheal disease.
3. If an outbreak pattern is detected in a specific zone, the DHO opens the **Healthcare Network Map**, identifies nearby clinics, and issues an operational advisory.

4. The DHO inspects the **Staff Administration Directory** to verify physician attendance and reassign relief personnel if a clinic is short-staffed.

Information Visible to the DHO

- Total district outpatient count, consultations completed, and prescriptions dispensed.
- Active clinic count, facility operational status, and system uptime.
- Aggregated disease patterns and top clinical diagnoses across the district.
- Facility-by-facility stock alerts indicating critical antibiotic or vaccine shortages.

Representative Screenshots

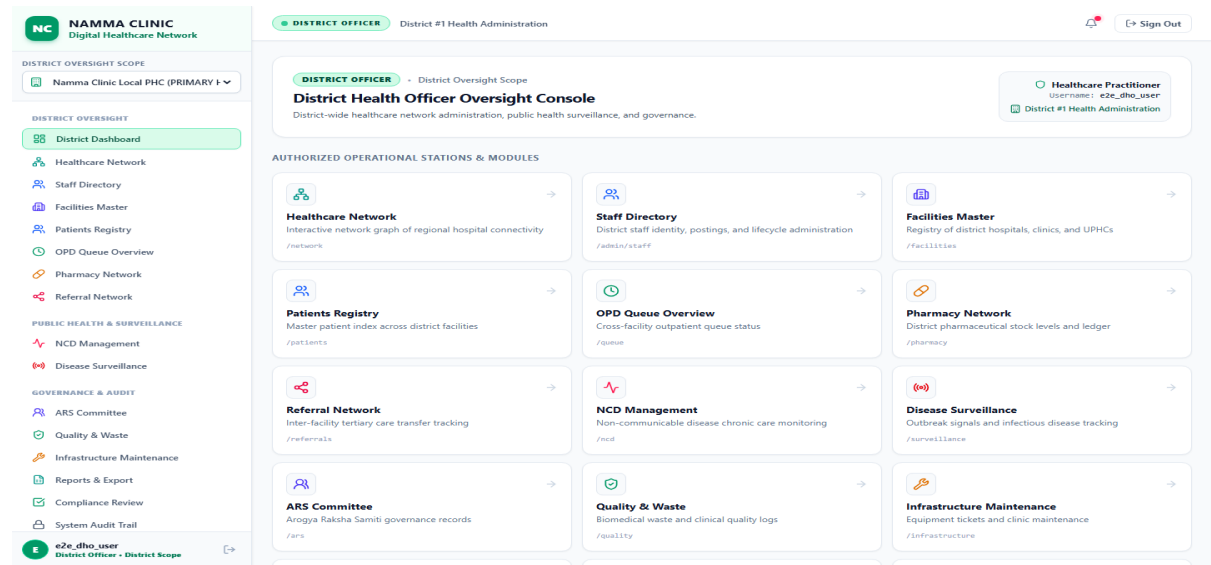


Figure 1 - District Health Officer Executive Dashboard - Displays total district attendance, active clinics, and district-wide healthcare delivery indicators.

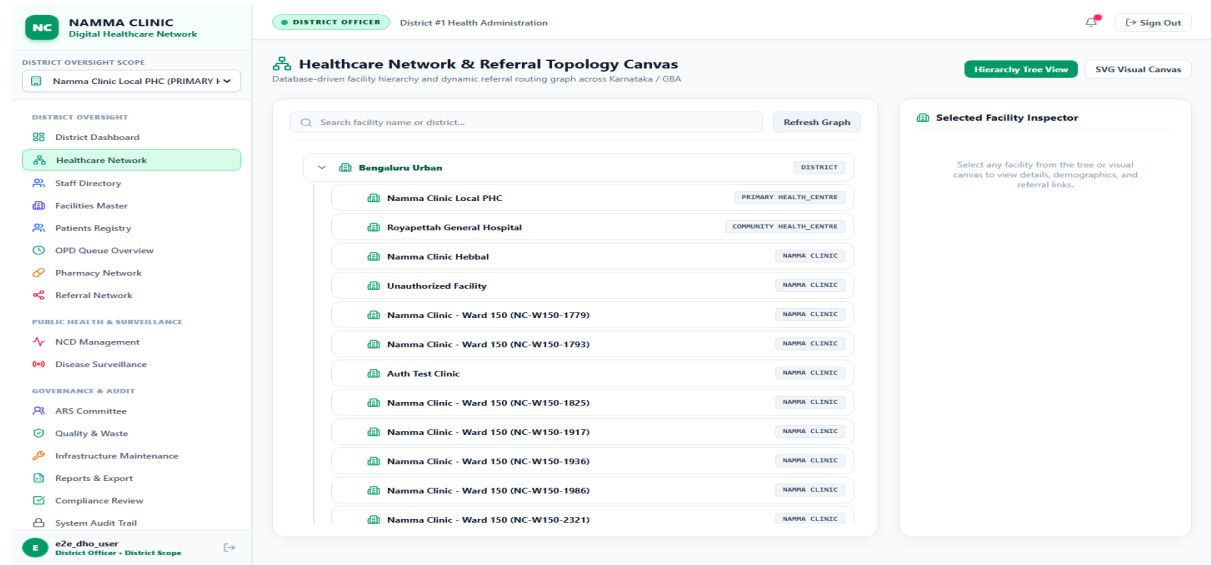


Figure 2 - Regional Healthcare Network Map - Interactive geographical view showing Primary Health Centers, Urban Clinics, and tertiary referral hospitals.

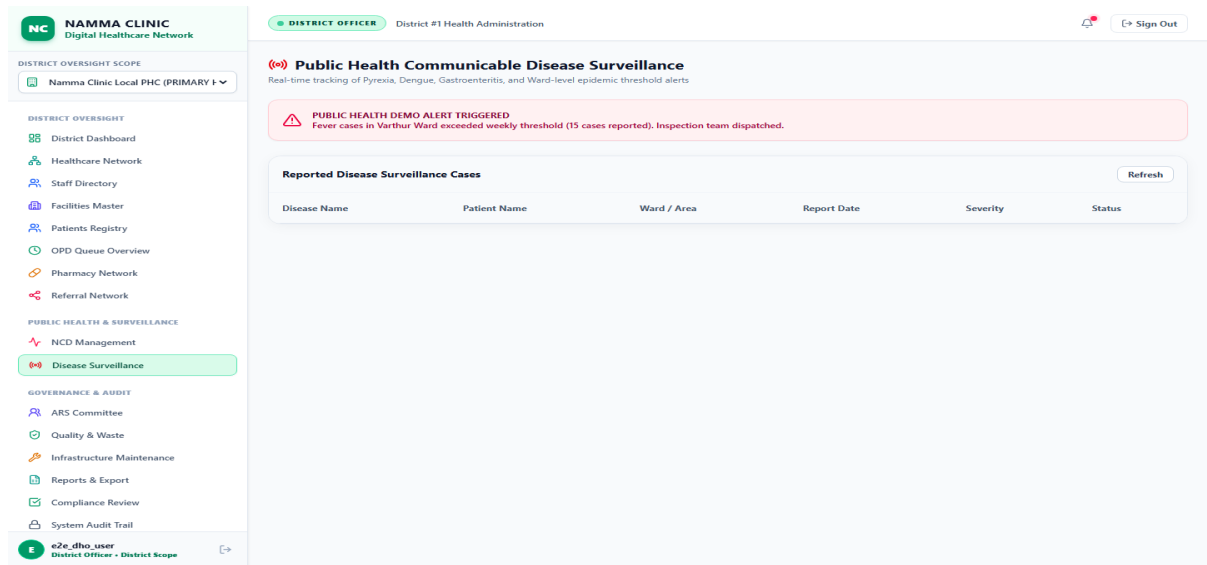


Figure 3 - District Public Health Surveillance Console - Epidemiological tracking console highlighting disease clusters and syndromic outbreak alerts.

2. Clinic / Hospital Administrator Workstation

Purpose

The **Clinic Administrator Console** equips the facility superintendent with day-to-day administrative oversight of clinic operations, staff scheduling, department setup, and operational compliance.

Core Responsibilities

- Supervise facility staff attendance, employee postings, and active duty assignments.
- Assign operational roles (Doctor, Nurse, Front Desk Officer, Pharmacist, Lab Technician).
- Oversee outpatient department flow and manage clinic overcrowding.
- Monitor critical facility stock depletion and equipment cold-chain temperatures.
- Review daily attendance logs and clinic operational reports.

Screens Available

- 1. **Clinic Admin Dashboard:** Overview of facility outpatient volume, queue length, and stock alerts.
- 2. **Staff Administration Console:** Employee directory, onboarding forms, and role assignment controls.
- 3. **Facility Operational Scope:** Clinic department management, operating hours, and room definitions.
- 4. **Facility Inventory Overview:** High-level stock balances and storekeeper purchase requisitions.
- 5. **Operational Reports:** Outpatient attendance summaries, morbidity distributions, and throughput.

Step-by-Step Workflow

- 1. The Administrator begins the day by checking the **Clinic Admin Dashboard** to verify that all clinical stations (Triage, Doctor Room, Lab, Pharmacy) are staffed.
- 2. If a newly hired nurse arrives, the Administrator opens **Staff Administration**, enters the nurse's details, assigns the NURSE role, and designates their primary clinic.
- 3. Mid-day, the Administrator monitors queue wait times to ensure patients move from triage to consultation within approved clinic standards.
- 4. At end of shift, the Administrator generates the **Daily Operational Report** for submission to district management.

Representative Screenshots

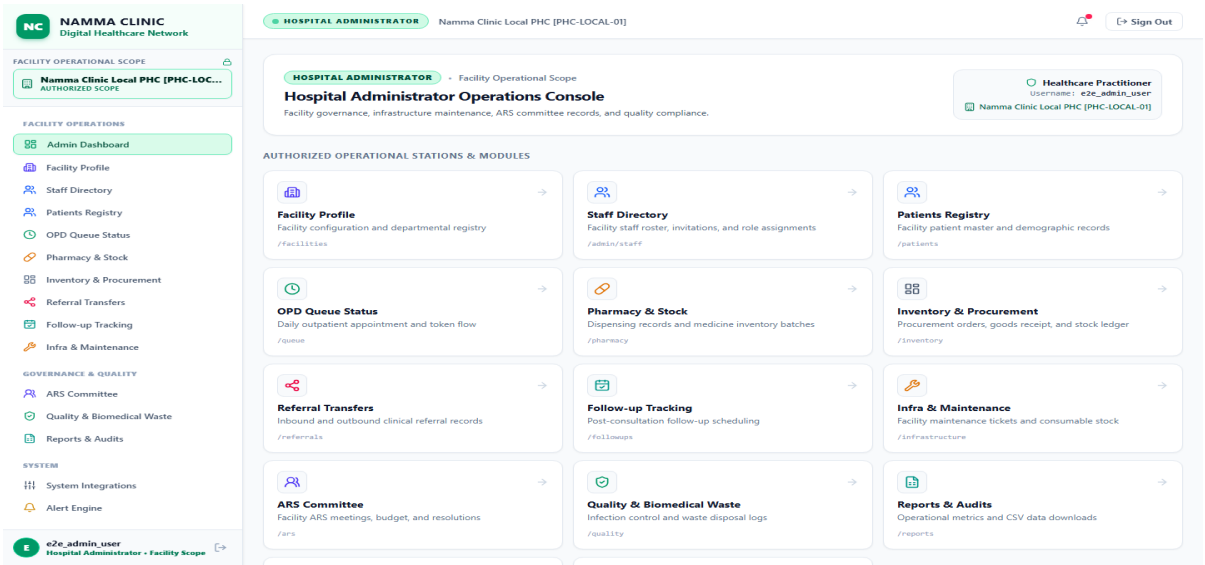


Figure 4 - Hospital Administrator Operational Dashboard - Facility-level operational KPIs showing waiting queue sizes, doctor availability, and daily attendance.

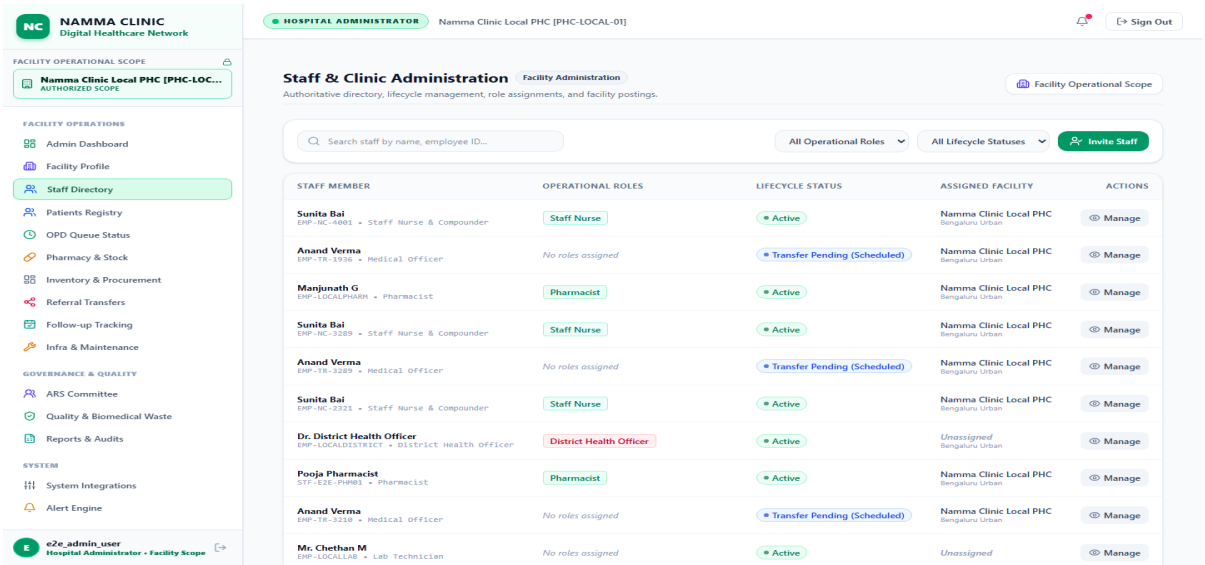


Figure 5 - Clinic Staff Administration & Role Allocation - Personnel management screen for inviting staff, assigning operational roles, and managing credentials.

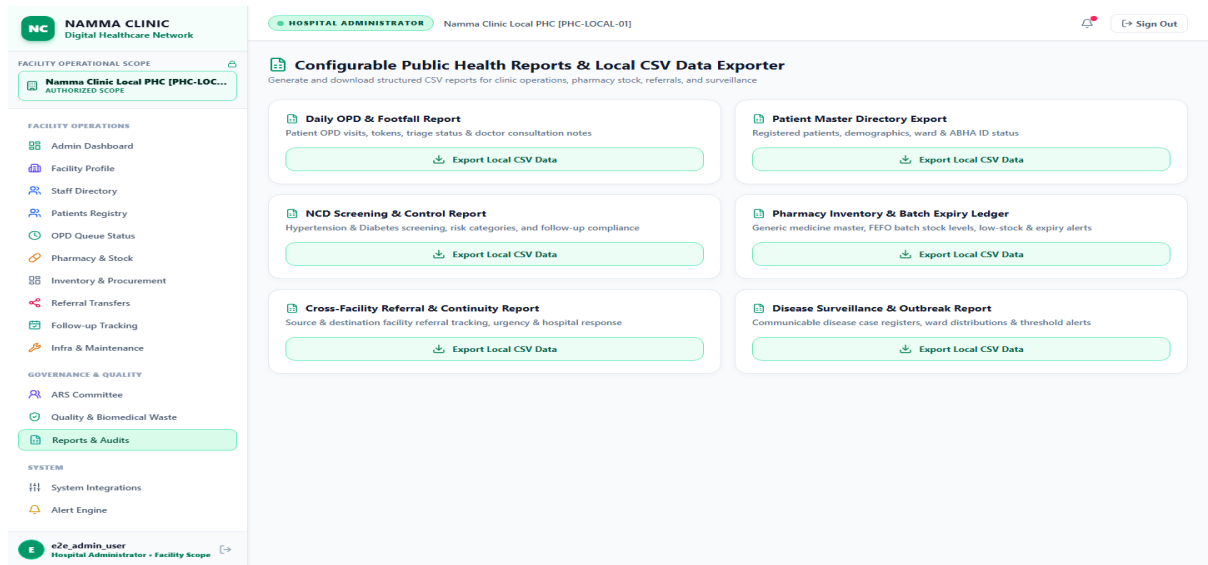


Figure 6 - Daily Clinic Reporting Engine - Comprehensive reporting module detailing patient demographics, diagnoses, and outpatient attendance summaries.

3. Medical Officer (Doctor) Workstation

Purpose

The **Doctor Consultation Workstation** provides clinicians with a clean, focused electronic medical record interface designed for rapid outpatient clinical decision-making.

Core Responsibilities

- Review waiting patient lists and call patients sequentially into the consultation room.
- Review pre-recorded vital signs, derived BMI, and triage acuity flags recorded by the nurse.
- Document presenting symptoms, clinical observations, and physical examination findings.
- Enter formal clinical diagnoses aligned with medical standards.
- Prescribe electronic medications with precise dosage, frequency, and duration instructions.
- Order diagnostic laboratory investigations and initiate specialty hospital referrals.

Screens Available

1. **Doctor Dashboard:** Summary of consultations completed today, waiting patients, and urgent alerts.
2. **Doctor Consultation Queue:** Prioritized list of triaged patients waiting for clinician examination.
3. **Clinical Consultation Form:** Integrated entry screen for clinical notes, diagnoses, and orders.
4. **Lab Results Review:** Diagnostic reports returned by the on-site lab technician.
5. **Referral Management:** Outgoing patient transfer documents for higher-tier specialty hospitals.

Step-by-Step Workflow

1. The Doctor opens the **Doctor Queue** and clicks **Call Next Patient**. The highest-priority waiting patient is displayed.
2. The Doctor reviews the nurse's triage findings - blood pressure, heart rate, temperature, SpO2, and nurse triage notes.
3. The Doctor interviews and examines the patient, recording clinical notes and selecting the confirmed or provisional diagnosis.
4. If testing is required, the Doctor checks the required diagnostic tests (e.g., Blood Sugar, Hemoglobin, Urine Routine), which automatically routes to the lab.
5. The Doctor selects medications from the clinic formulary, specifying instructions (e.g., *1 tablet twice daily after food for 5 days*).
6. Clicking **Complete Consultation** automatically updates the patient's record, dispatches the e-prescription to the dispensary, and calls the next patient.

Representative Screenshots

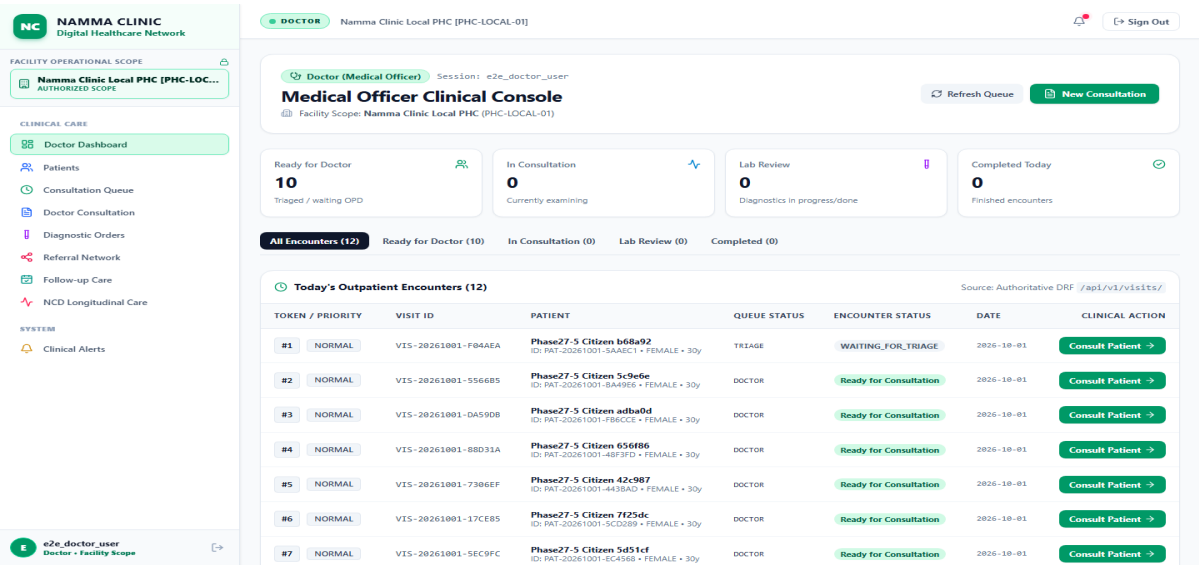


Figure 7 - Doctor Workstation Consultation Dashboard - Clinician home screen showing waiting consultation queue, completed patient encounters, and urgent clinical flags.

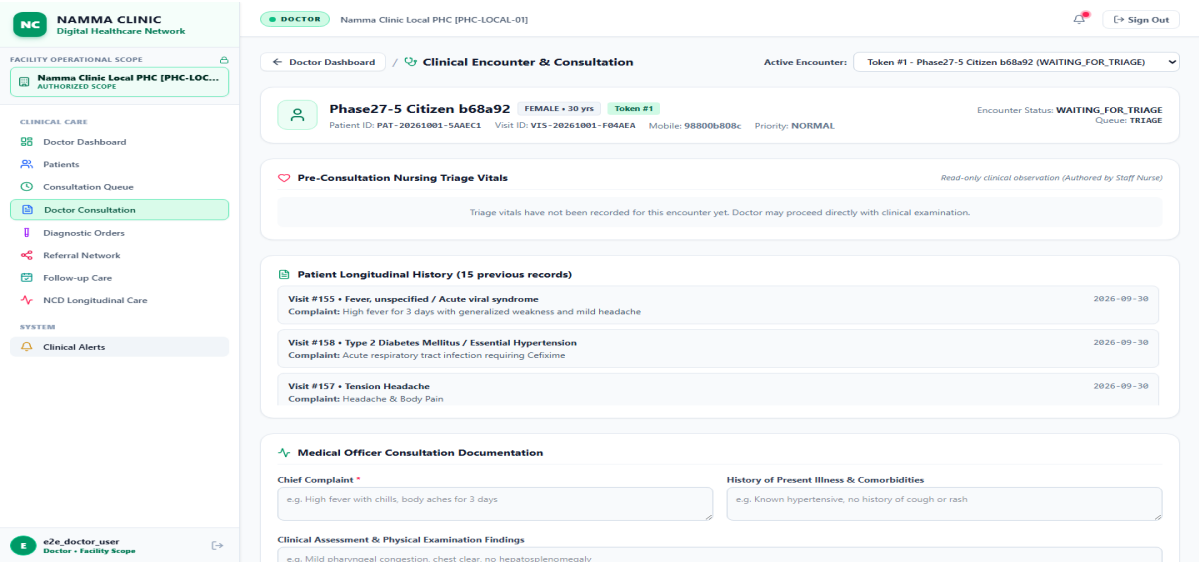


Figure 8 - Clinical Examination & Diagnosis Form - Consultation documentation screen displaying nurse triage vitals, clinical note inputs, and diagnostic ordering fields.

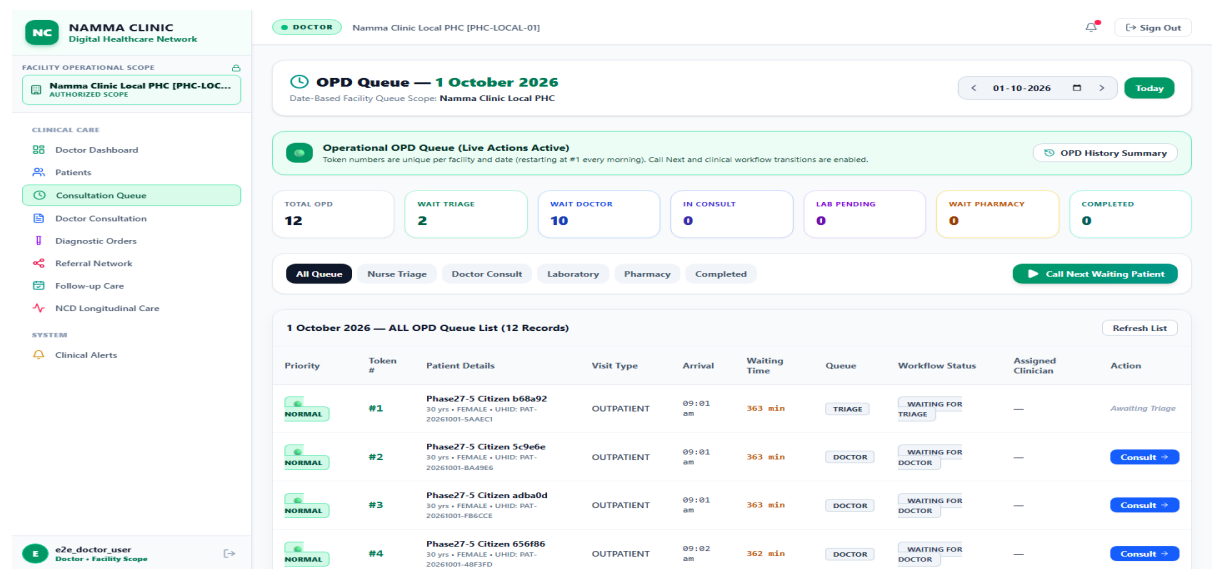


Figure 9 - Doctor Outpatient Queue View - Prioritized waiting list displaying patient token numbers, triage acuity colors, and arrival times.

4. Staff Nurse Workstation

Purpose

The **Nurse Triage & Clinical Workstation** enables nurses to prioritize incoming walk-in patients through standardized vital sign measurements, acuity assessments, and community wellness tracking.

Core Responsibilities

- Call waiting walk-in patients from reception into the nursing station.
- Measure and document vital signs: Systolic/Diastolic Blood Pressure, Pulse, Temperature, SpO2, Respiratory Rate, Height, and Weight.
- Review automated BMI calculations and evaluate vital abnormalities.
- Assign clinical acuity scores (IMMEDIATE, VERY_URGENT, URGENT, STANDARD, NON_URGENT).
- Screen citizens for chronic Non-Communicable Diseases (Hypertension and Diabetes).
- Manage patient follow-up recalls, immunization drives, and community health camps.

Screens Available

1. **Nurse Dashboard:** Overview of waiting walk-ins, triage completion rates, and vitals alerts.
2. **Triage Queue:** Real-time list of newly registered patients waiting for nursing assessment.
3. **Vitals Recording Form:** Clean clinical interface for entering physiological measurements.
4. **NCD Registry:** Screening and tracking panel for chronic hypertensive and diabetic patients.
5. **Community Outreach & Wellness:** Logs for slum health visits, maternal counseling, and health education.

Step-by-Step Workflow

1. The Nurse views the **Triage Queue** and calls the next patient token issued by reception.
2. The Nurse measures the patient's vital signs and enters the numbers into the **Vitals Form**.
3. The platform computes the Body Mass Index automatically and highlights any abnormal values in amber or red.
4. The Nurse records presenting symptoms and assigns an acuity level based on clinical evaluation.
5. Upon saving, the patient's status transitions to TRIAGED and appears immediately in the Doctor's consultation queue.

Representative Screenshots

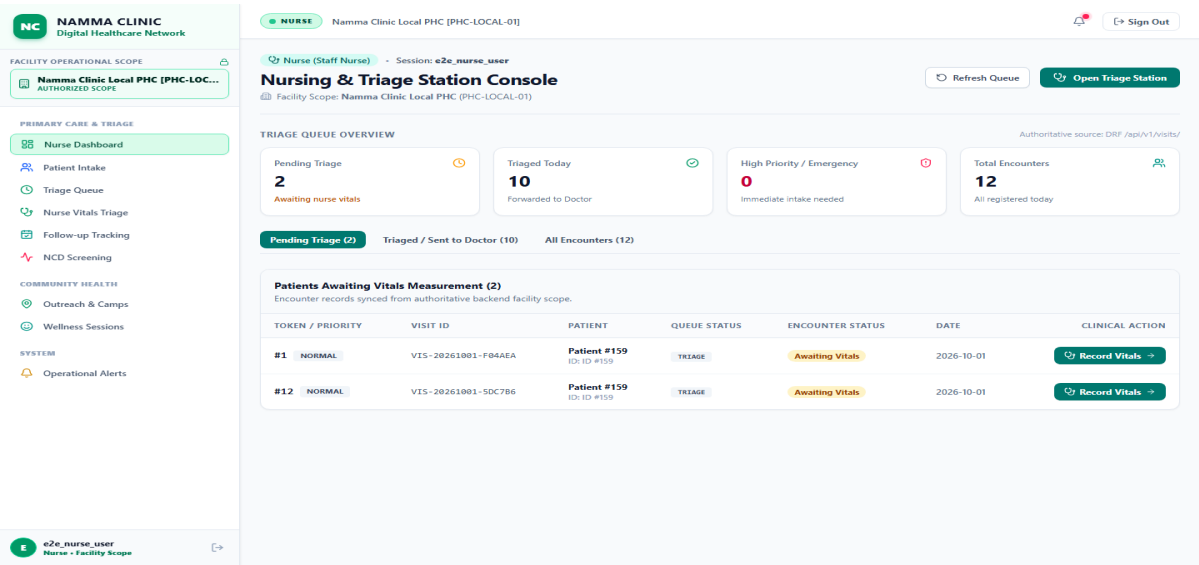


Figure 10 - Staff Nurse Triage Dashboard - Nursing home screen displaying walk-in intake numbers, pending vitals assessments, and chronic care alerts.

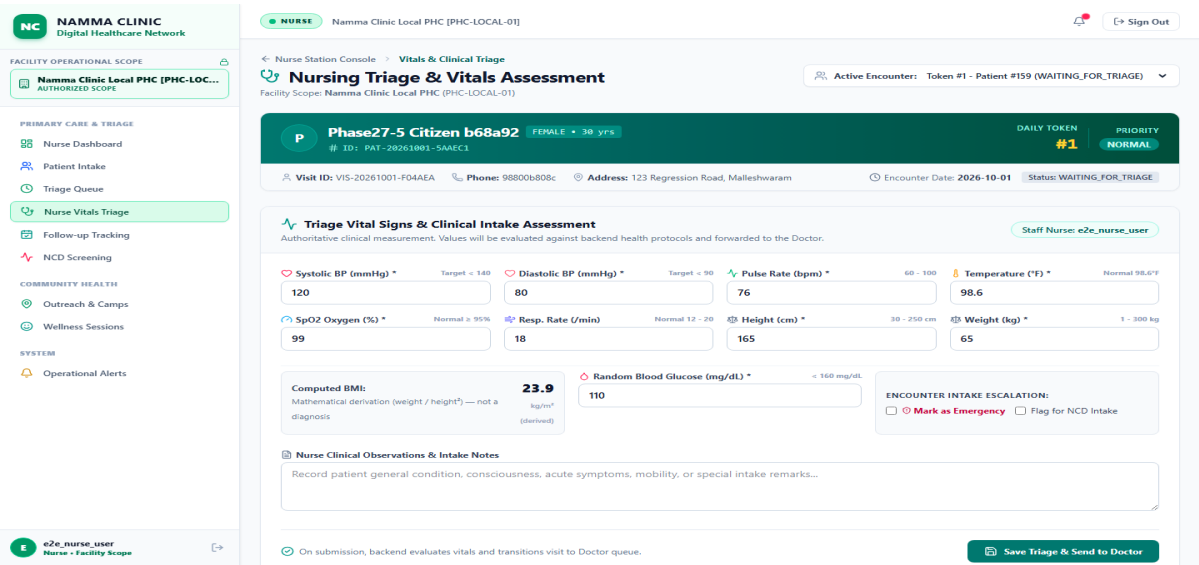


Figure 11 - Clinical Vitals Assessment Form - Standardized vitals capture form with automated BMI computation and clinical acuity prioritization.

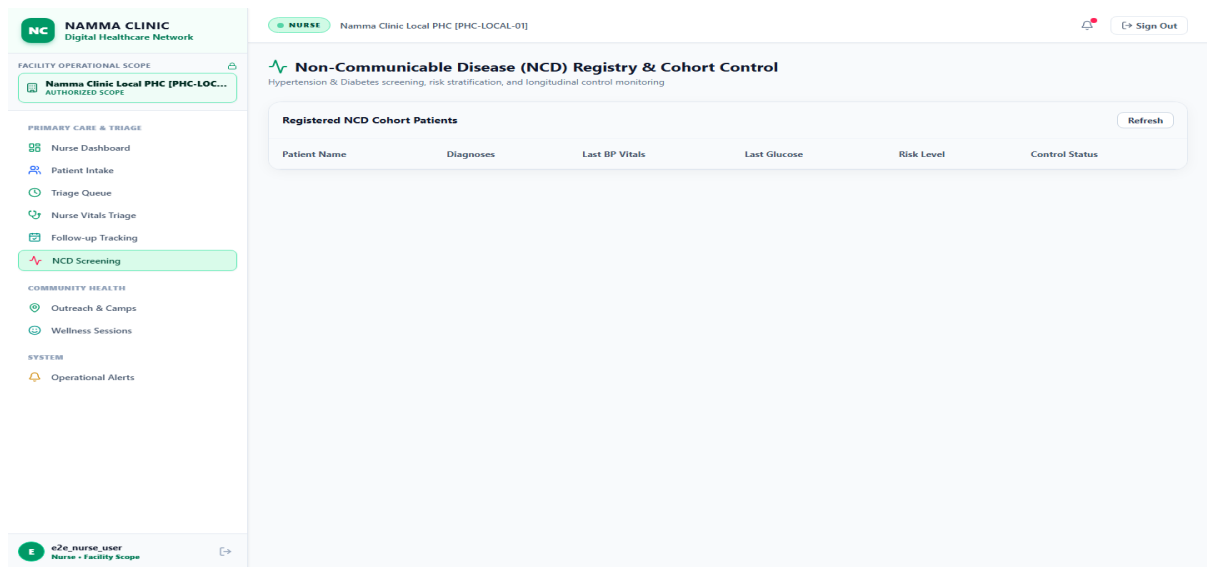


Figure 12 - Non-Communicable Disease Screening Panel - Chronic disease registry tracking blood pressure readings, glucose monitoring, and recall dates.

5. Front Desk Officer Workstation

Purpose

The **Front Desk Officer Workstation** handles the initial reception, demographic verification, new citizen onboarding, and sequential OPD token dispatch.

Core Responsibilities

- Greet citizens arriving at the clinic reception desk.
- Search the Master Patient Index by mobile phone number, full name, or health ID.
- Register new citizens with contact details, residential address, and vulnerability group.
- Generate sequential daily OPD tokens (e.g., T-001, T-002) and assign patients to clinic departments.
- Void or cancel tokens issued in error before nursing triage commences.
- **Privacy Boundary:** Front Desk Officers cannot view clinical consultations, medical notes, diagnoses, or prescriptions.

Screens Available

- 1. **Front Desk Console:** Reception overview with daily token statistics and rapid search.
- 2. **Patient Directory:** Master citizen demographic registry.
- 3. **New Patient Registration Modal:** Intake form for demographic and contact information.
- 4. **OPD Queue View:** Facility-wide token status monitor.

Step-by-Step Workflow

- 1. The Front Desk Officer asks the patient for their mobile number or name and searches the registry.
- 2. If the patient exists, the officer verifies their address and clicks **Issue OPD Token**.
- 3. If the patient is new, the officer clicks **Register Patient**, enters their demographics, and clicks **Register & Issue Token**.
- 4. A daily sequential token is minted, and the patient is directed to the nursing triage waiting area.

Representative Screenshots

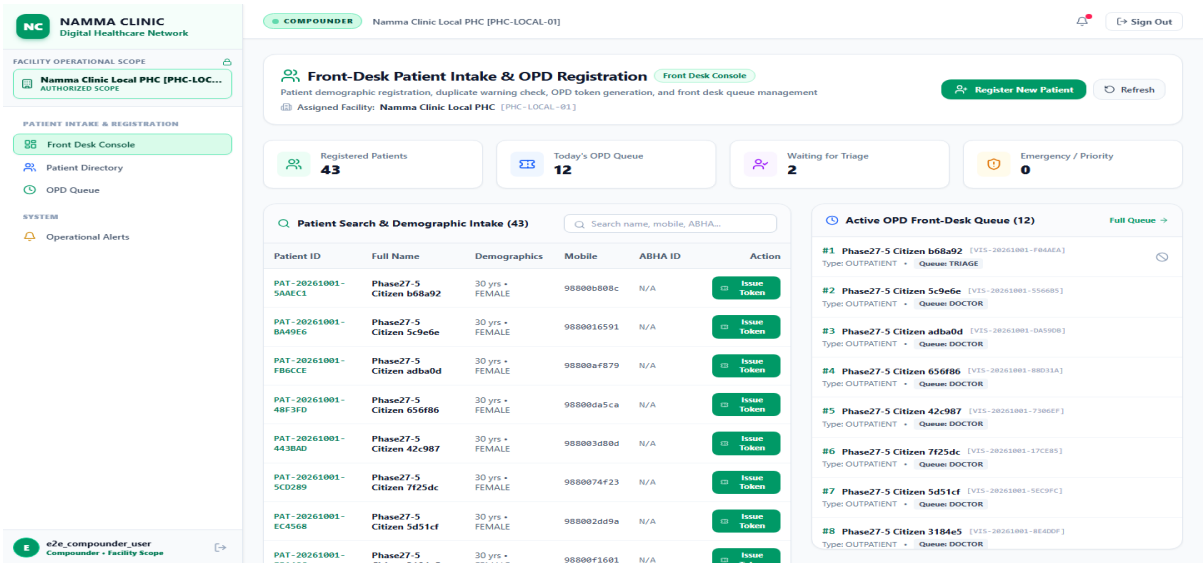


Figure 13 - Front Desk Reception Console - Intake workstation showing daily registration counts, search bar, and token queue statistics.

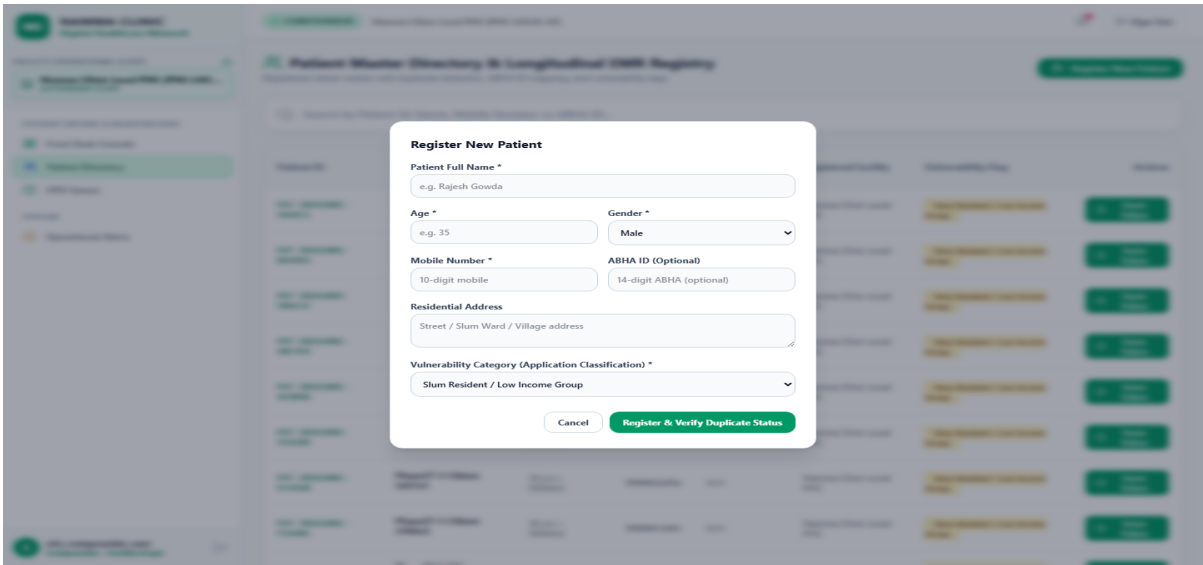


Figure 14 - Citizen Intake Registration Modal - Clean registration dialog capturing demographic details, emergency contacts, and neighborhood vulnerability.

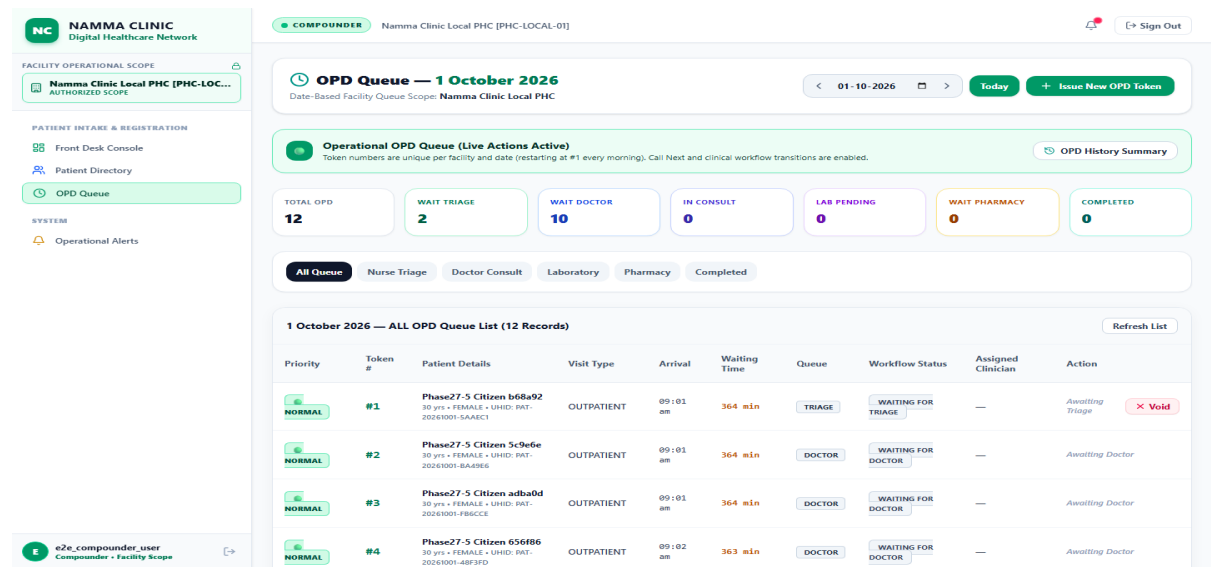


Figure 15 - Outpatient Token Queue Monitor - Real-time facility waiting queue displaying waiting tokens, patient identifiers, and priority designations.

6. Diagnostic Lab Technician Workstation

Purpose

The **Diagnostic Laboratory Workstation** enables laboratory technicians to manage test orders, track sample collection, input numeric test results, and publish verified diagnostic reports.

Core Responsibilities

- Monitor incoming laboratory requisitions ordered by clinic doctors in real time.
- Verify patient identity and record specimen collection (Blood, Urine, Sputum, etc.).
- Perform diagnostic investigations and enter numeric parameters.
- Verify results against standard biological reference ranges and publish verified reports.

Screens Available

1. **Lab Dashboard:** Summary of pending test orders, samples collected, and reports completed today.
2. **Diagnostic Test Queue:** List of patients waiting for sample collection or test processing.
3. **Investigation Entry Screen:** Structured data form for entering diagnostic test parameters.

Step-by-Step Workflow

1. When a doctor orders tests, the order appears immediately on the **Lab Queue**.
2. When the patient arrives at the lab, the technician clicks **Collect Specimen**, confirming sample receipt.
3. After completing the bench analysis, the technician opens the patient's order and enters the numeric values.
4. The system validates the input against biological reference ranges and highlights abnormal results.
5. The technician clicks **Publish Report**, which instantly updates the doctor's workstation.

Representative Screenshots

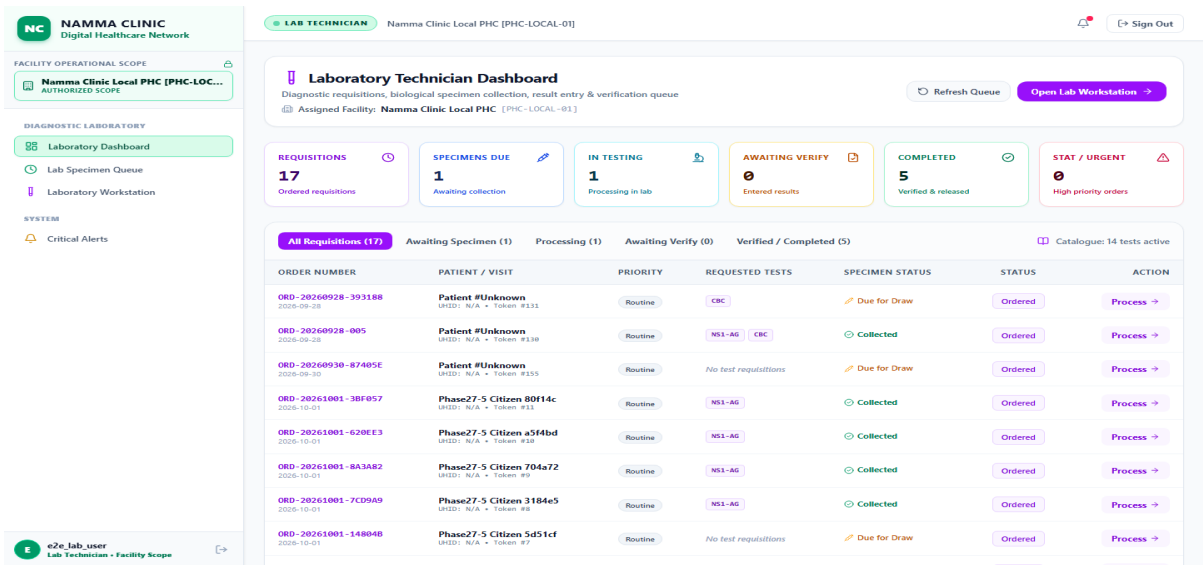


Figure 16 - Diagnostic Laboratory Dashboard - Lab home screen showing pending investigations, collected samples, and completed diagnostic reports.

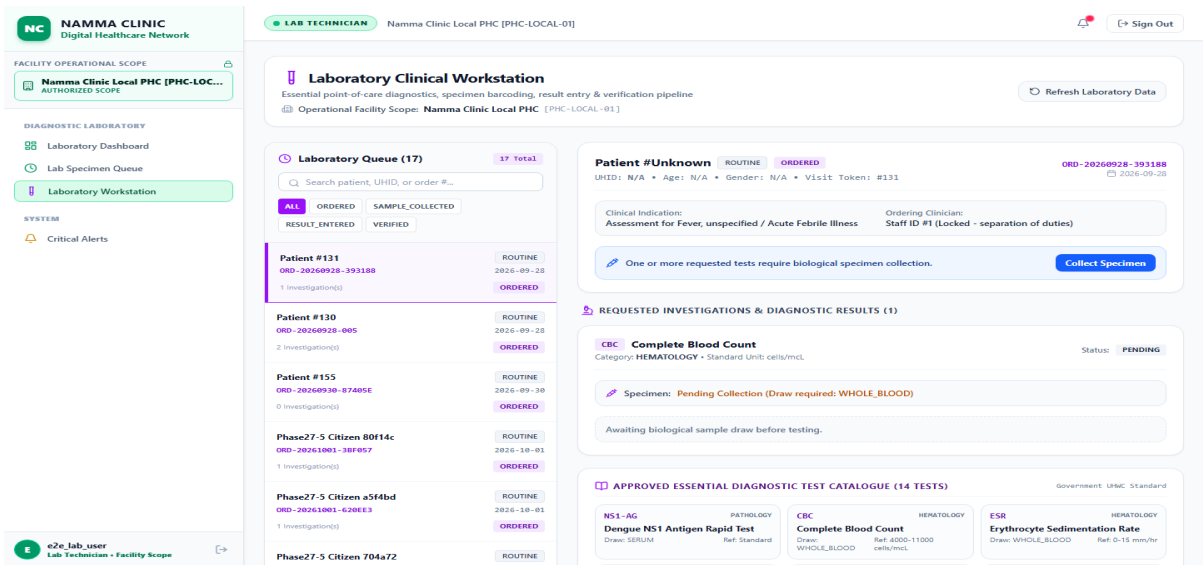


Figure 17 - Diagnostic Investigation Test Queue - Specimen intake list allowing technicians to verify samples and publish clinical parameters.

7. Pharmacist Workstation

Purpose

The **Pharmacy Workstation** allows pharmacists to review doctor prescriptions, select First-Expired, First-Out (FEFO) medicine batches, and dispense medications safely.

Core Responsibilities

- Review electronic prescriptions issued by clinic doctors.
- Check medication availability, dosage forms, and prescribed durations.
- Verify system-recommended First-Expired, First-Out (FEFO) drug batches.
- Place invalid prescriptions on hold with clinical explanation notes.
- Dispense medications, record patient counseling, and automatically decrement facility pharmacy stock.

Screens Available

1. **Pharmacy Dashboard:** Active prescription queue, dispensing metrics, and stock alerts.
2. **Dispensing Workstation:** Detailed prescription verification and batch selection screen.

- 3. **Pharmacy Drug Stock View:** Available medication inventory within the dispensary.
- 4. **Storage & Infrastructure:** Storage condition logs (e.g., refrigerator cold-chain monitoring).

Step-by-Step Workflow

- 1. The Pharmacist views the **Pharmacy Dashboard** as new electronic prescriptions arrive.
- 2. The Pharmacist calls the patient token and opens the electronic prescription.
- 3. For each prescribed drug, the platform displays the prescribed quantity and automatically highlights the earliest-expiring active batch in stock.
- 4. The Pharmacist verifies the physical medicine against the screen and clicks **Dispense Prescription**.
- 5. The platform logs the dispensation, prints or displays dispensing instructions, and decrements stock in real time.

Representative Screenshots

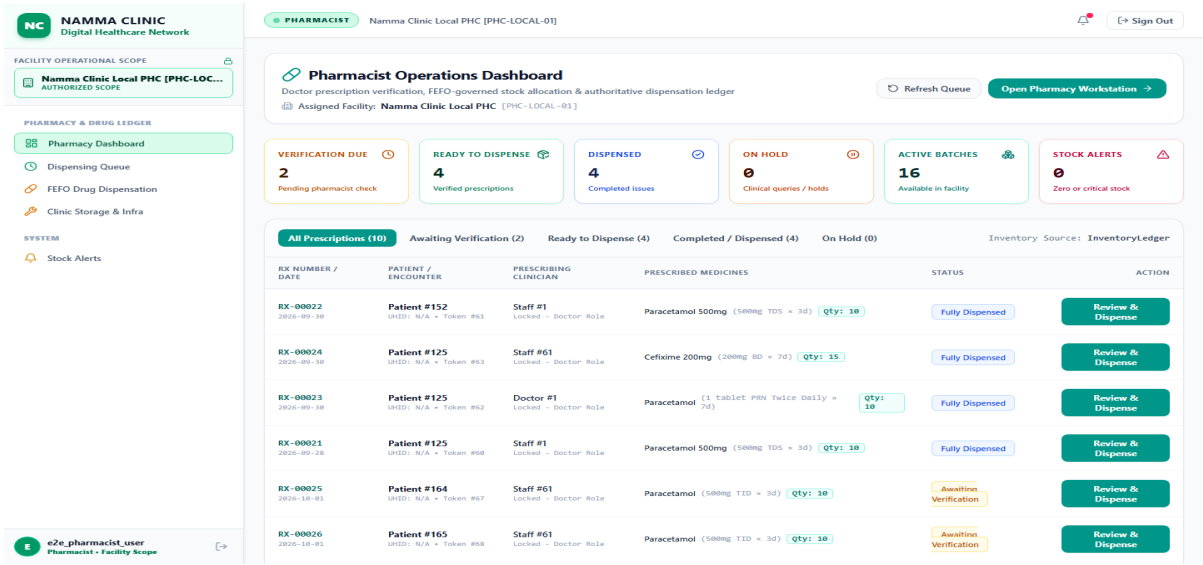


Figure 18 - Pharmacy Dispensing Dashboard - Pharmacist home screen tracking pending prescriptions, dispensed tokens, and low-stock warnings.

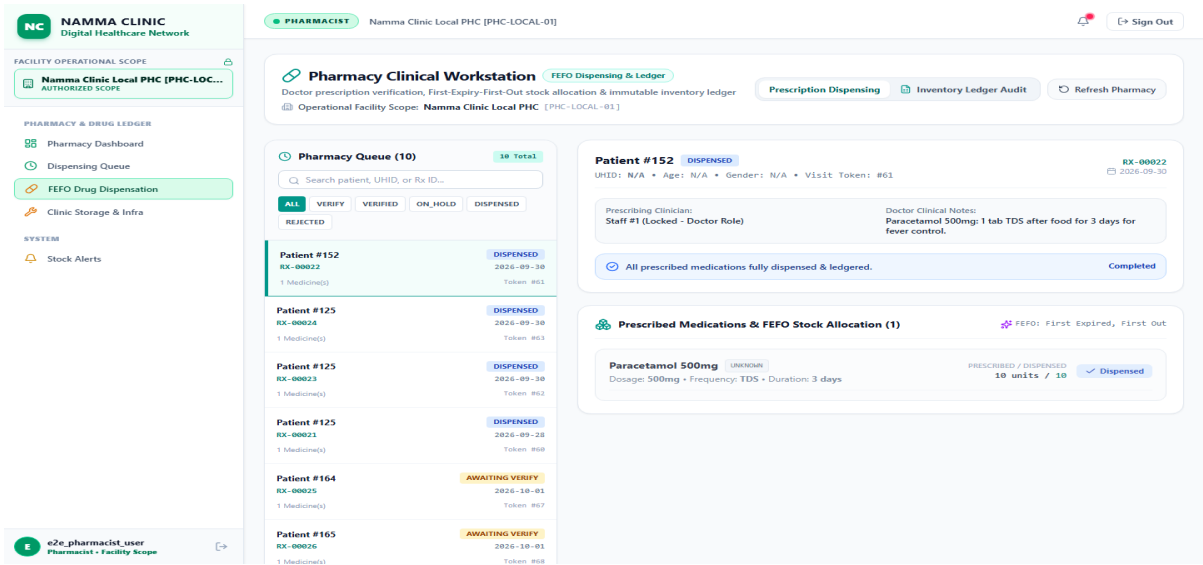


Figure 19 - Electronic Prescription Verification & Dispensing - Prescription verification table showing doctor orders, recommended FEFO batches, and dispense controls.

8. Inventory Officer Workstation

Purpose

The **Inventory & Procurement Console** enables storekeepers to manage central clinic warehouse stock, raise vendor purchase orders, receive shipments with Goods Receipt Notes (GRN), and perform stock reconciliations.

Core Responsibilities

- Monitor bulk warehouse stock balances and minimum reserve thresholds.
- Create formal Purchase Orders (PO) to approved pharmaceutical suppliers.
- Inspect incoming shipments and process Goods Receipt Notes with batch numbers and expiry dates.
- Perform physical inventory audits and log signed adjustments.
- **Separation of Duties:** Inventory Officers manage warehouse receiving; they cannot dispense medications directly to patients.

Screens Available

1. **Inventory Workstation:** Bulk warehouse balances, batch listings, and expiry dates.
2. **Procurement & Purchase Orders:** Purchase order creation, approval status, and supplier records.
3. **Goods Receipt (GRN) Module:** Shipment intake dialog for verifying delivered quantities.
4. **Inventory Movement Ledger:** Immutable chronological history of all stock additions and transfers.

Step-by-Step Workflow

1. The Inventory Officer checks the **Inventory Workstation** to review items approaching reorder levels.
2. To order supplies, the officer opens **Purchase Orders**, selects the supplier, adds medicine items with requested quantities, and submits the order.
3. When the shipment arrives, the officer opens the PO and clicks **Create Goods Receipt Note (GRN)**.
4. The officer enters the manufacturer's batch numbers, manufacturing dates, expiry dates, and received quantities.
5. Saving the GRN automatically updates the warehouse stock ledger and makes the batches available for clinic dispensing.

Representative Screenshots

MEDICINE	BATCH #	EXPIRY DATE	AVAILABLE STOCK	SUPPLIER / VENDOR	STATUS	ACTIONS
Paracetamol Dolo	EZE-REC-BATCH-01	2027-06-30	303	Apex Pharma Distributors	AVAILABLE	Reconcile
Paracetamol 500mg	BATCH-LOC-PCR01	2027-06-30	480	—	AVAILABLE	Reconcile
Amoxicillin 500mg	BATCH-LOC-APX01	2027-08-31	300	—	AVAILABLE	Reconcile
Paracetamol Dolo	EZE-GRN-BATCH-6589	2027-09-30	100	Apex Pharma Distributors	AVAILABLE	Reconcile
Paracetamol Dolo	EZE-GRN-BATCH-7942	2027-09-30	100	Apex Pharma Distributors	AVAILABLE	Reconcile

Figure 20 - Inventory Workstation & Batch Ledger - Real-time facility inventory ledger displaying batch quantities, expiry dates, and unit prices.

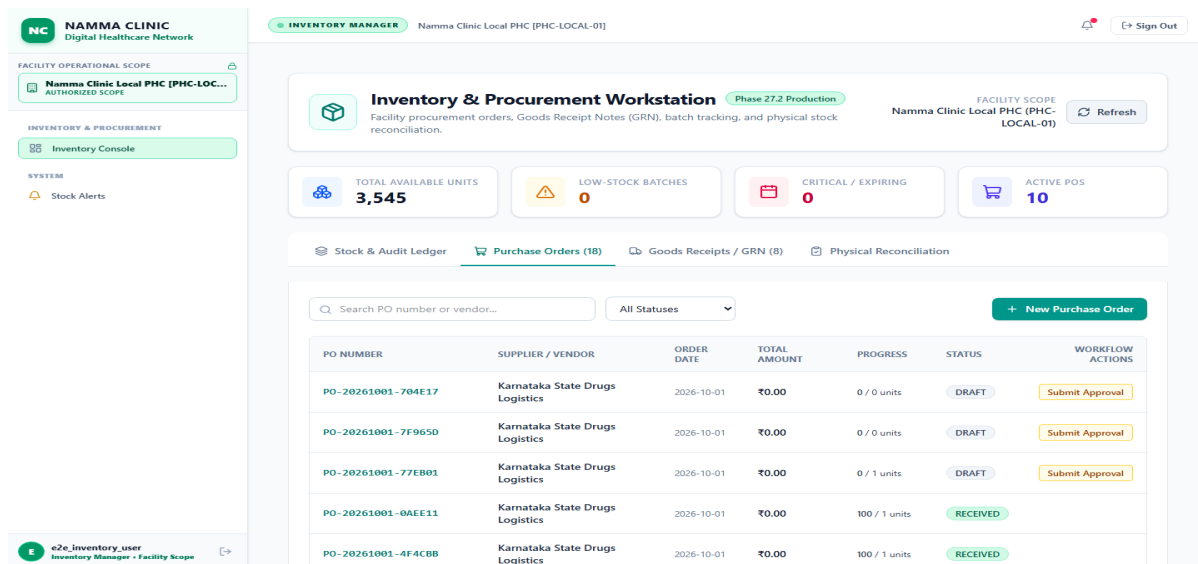


Figure 21 - Procurement Purchase Orders Tab - Procurement dashboard showing active vendor purchase orders, delivery statuses, and creation modal controls.

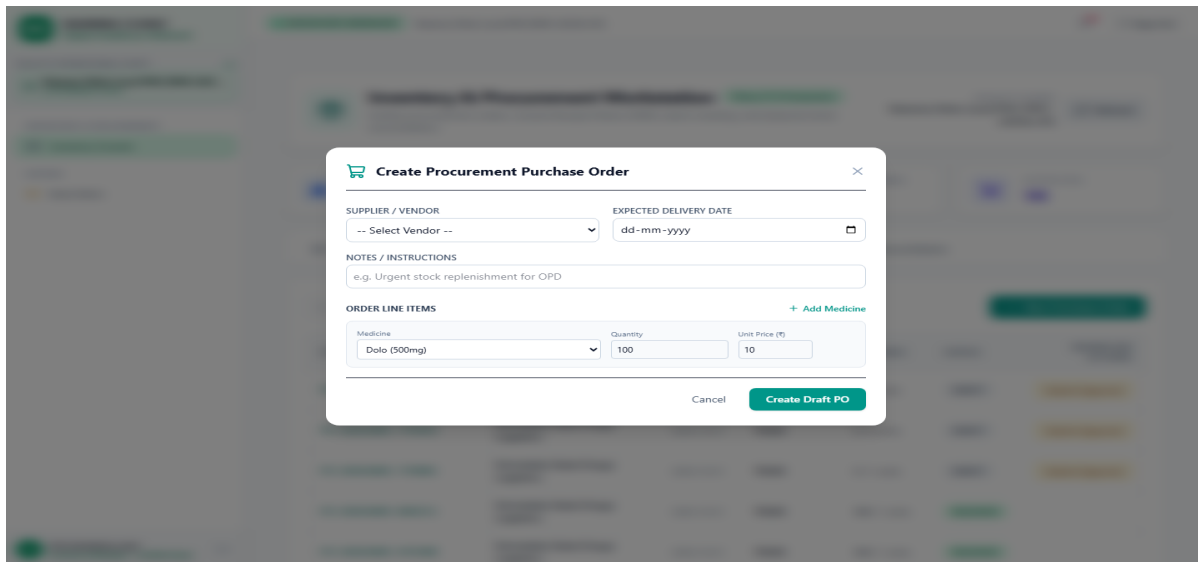


Figure 22 - Purchase Order Generation Modal - Supplier order creation dialog specifying ordered drug quantities, unit costs, and delivery terms.

PART C - DUAL-ROLE OPERATIONS (SMALL-CLINIC RULE)

Primary healthcare facilities across urban and rural sectors often operate with lean personnel. A neighborhood clinic or health outpost may have only one nurse or one pharmacist on duty during a shift.

Rather than compromising administrative controls by creating informal or combined roles (which introduce severe security and auditing risks), the Namma Clinic Platform supports the **Small-Clinic Rule**: an administrator can grant **two distinct, standard operational roles** to a single verified staff member. The employee carries out each duty under its dedicated operational rules, distinct screens, and audit tracking.

1. Nurse + Front Desk Officer Dual Role

In a neighborhood clinic without a dedicated reception clerk, the Staff Nurse holds two distinct role assignments: NURSE and FRONT DESK OFFICER.

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- Searches citizen registry & enters new walk-ins

• Issues daily sequential OPD token (e.g., T-101)

▼

STEP 2: CLINICAL TRIAGE (Nursing Duty)

• Nurse moves to clinical triage station

• Calls token T-101 into triage workstation

• Measures vitals, calculates BMI & assigns acuity

• Hands off triaged patient to doctor consultation

Operational Separation of Responsibilities

- **Strict Privacy Boundaries:** When greeting citizens and issuing tokens at reception, the nurse uses front-desk screens and cannot see confidential clinical consultation notes or diagnostic history of past visits. This prevents accidental exposure of sensitive diagnoses in the public reception area.
- **Clinical Triage Integrity:** When measuring physiological vitals and assessing acute symptoms, the nurse switches to the clinical workstation. The vital signs immediately appear on the Doctor's queue with color-coded acuity flags.
- **Audit Trail Accountability:** Every token generated is logged under the FRONT_DESK_OFFICER role permission, while every blood pressure reading is logged under the NURSE role permission.
- **Flexible Clinic Scaling:** When patient volume increases and a dedicated front desk officer is hired, the administrator simply revokes the FRONT_DESK_OFFICER assignment from the nurse's profile. The nurse immediately returns to standard single-role nursing without data loss or administrative disruption.

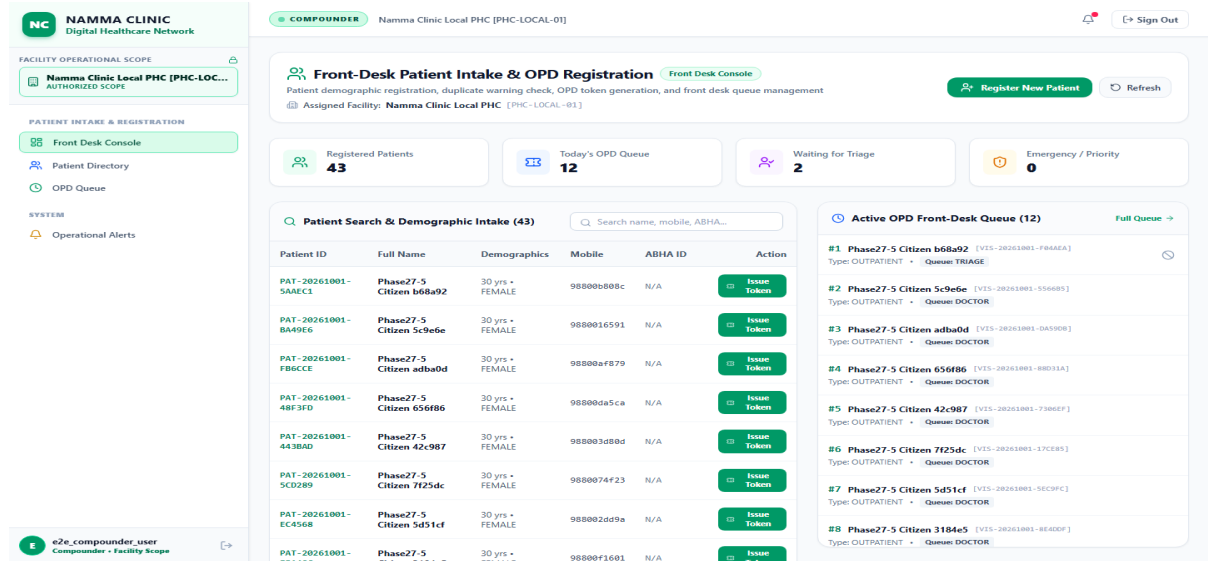


Figure 23 - Step 1: Reception Intake via Front Desk Duty - The nurse accesses the reception console to look up walk-in citizens, verify demographic profiles, and issue sequential OPD tokens.

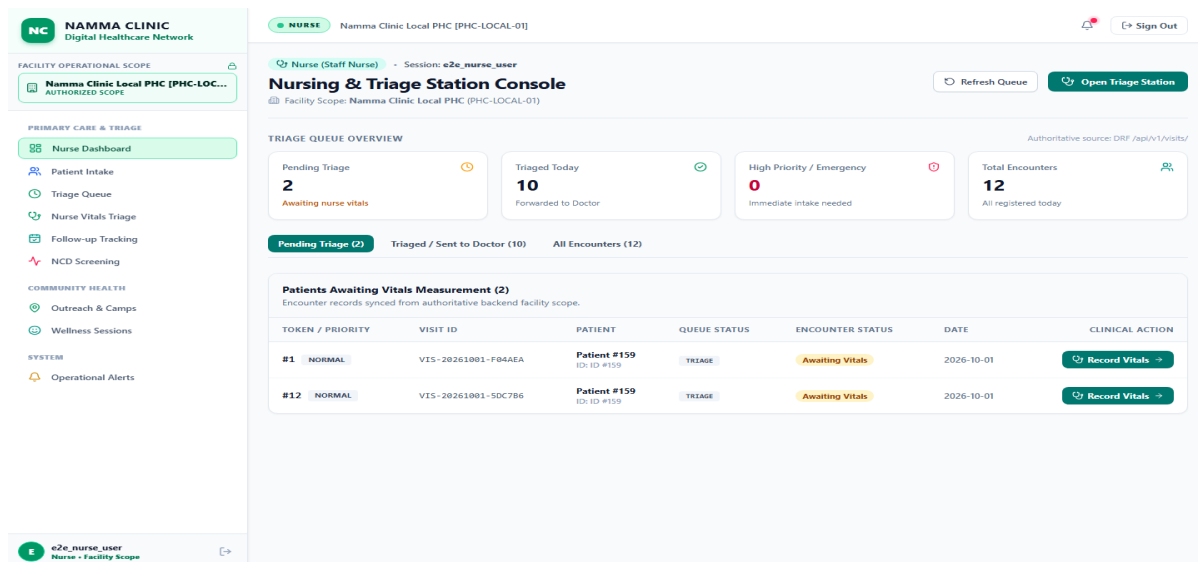
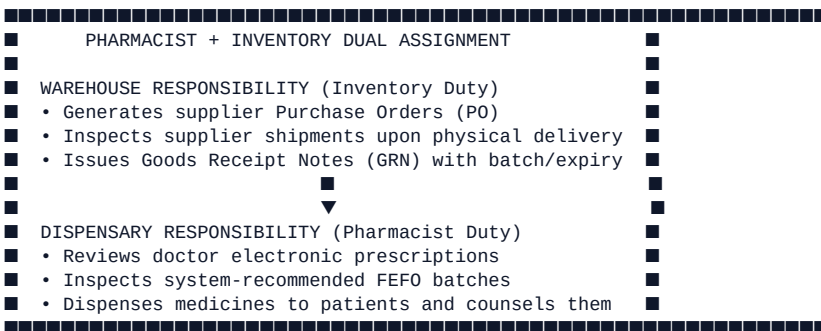


Figure 24 - Step 2: Clinical Vitals Triage via Nurse Duty - Switching to the nursing console, the nurse calls the token into the triage station to document vital signs, compute BMI, and assign priority flags.

2. Pharmacist + Inventory Officer Dual Role

In clinics where a single pharmacy professional manages both the warehouse drug stockroom and the outpatient dispensing counter, the employee holds both PHARMACIST and INVENTORY role assignments.



Operational Separation of Responsibilities

- **Inventory Intake Before Dispensing:** The professional cannot dispense medicines directly out of shipping boxes. When a shipment arrives from the district warehouse, they must log into the Inventory workstation and record a verified Goods Receipt Note (GRN) capturing batch numbers, expiry dates, and unit quantities.
- **Automated FEFO Enforcement:** When dispensing at the pharmacy counter, the system automatically allocates the earliest-expiring active batch first, preventing older stock from expiring unnoticed on back shelves.
- **Strict Double-Entry Accountability:** Dispensing deductions cannot be fabricated; every medicine issued must be tied to a valid Doctor consultation and electronic prescription.
- **Clean Responsibility Handoff:** If the facility expands and hires a separate warehouse storekeeper, the administrator reassigns the INVENTORY role to the new hire. The pharmacist retains dispensing responsibilities with zero operational downtime.

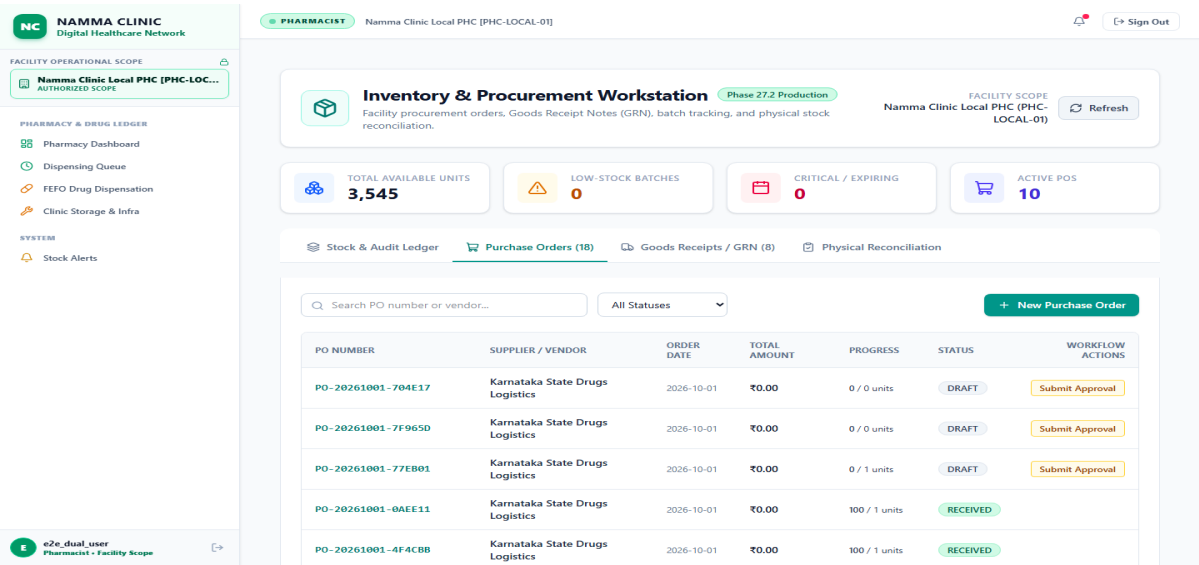


Figure 25 - Step 1: Warehouse Stock Management via Inventory Duty - The dual-role officer monitors clinic stock levels, creates purchase orders, and logs verified Goods Receipt Notes into the store ledger.

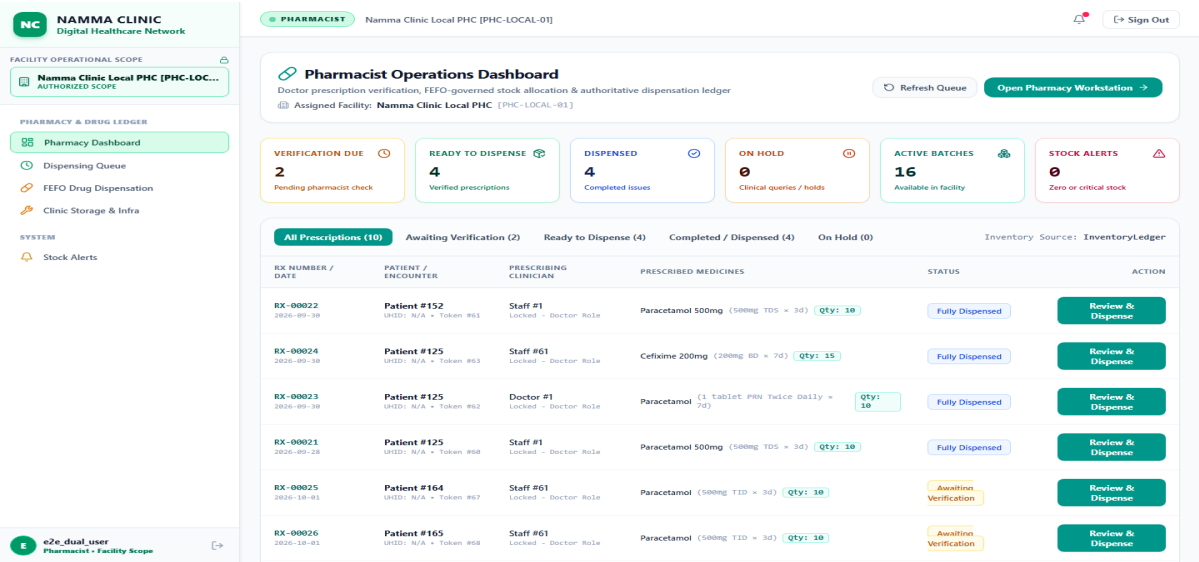
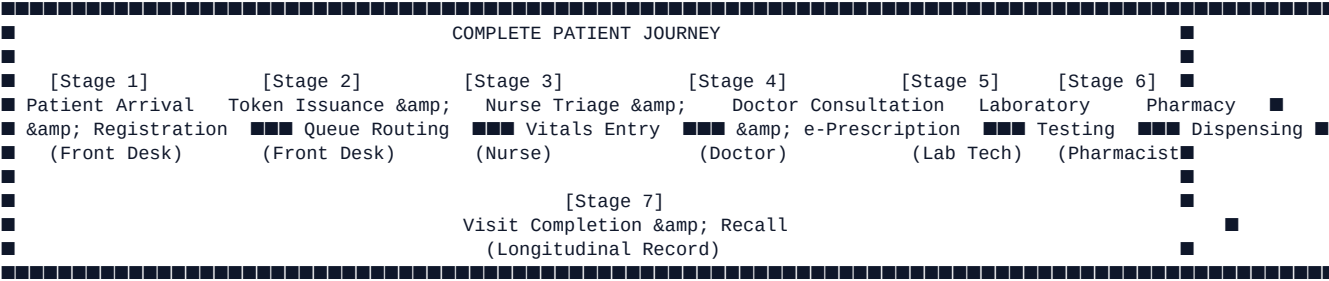


Figure 26 - Step 2: Outpatient Prescription Dispensing via Pharmacist Duty - Switching to the dispensary view, the officer validates doctor e-prescriptions, confirms FEFO batch allocations, and dispenses medications.

PART D - COMPLETE PATIENT JOURNEY

Every patient visit follows a standardized, chronological clinical lifecycle. This section traces a citizen through all seven operational stages, detailing who acts, what data is recorded, and what happens next.



Stage-by-Stage Operational Walkthrough

Stage 1: Patient Arrival & Demographic Search

- **Who Performs It:** Front Desk Officer (or Staff Nurse under dual-role rule).
- **What They Do:** Greets citizen, searches phone number or name in the central patient registry. If new, opens the registration modal and captures primary demographics (Name, Age, Gender, Mobile, Address, Emergency Contact).
- **Information Captured:** National health identifier, full name, age, biological sex, phone number, residential neighborhood, and guardian details.
- **What Happens Next:** The patient's permanent record is saved and opened for outpatient token creation.
- **What the Next User Sees:** The triage nurse sees the patient appear on the incoming triage waiting list.
- **Why it Matters:** Eliminates duplicate medical charts, establishes verifiable citizen identity, and protects clinical privacy by restricting front desk staff from viewing past medical notes.
- **Management Monitoring:** Tracks total daily walk-in registrations, demographic completeness rates, and peak arrival hours.

Figure 27 - Citizen Demographic Registration Screen - Front desk interface capturing essential citizen demographic details (name, age, gender, mobile, address) to establish a permanent clinic record without exposing medical history.

Stage 2: Token Issuance & Queue Placement

- **Who Performs It:** Front Desk Officer.
- **What They Do:** Generates a daily sequential outpatient (OPD) token (e.g., Token #T-104) and directs the patient to the nursing triage waiting area.
- **Information Captured:** Visit timestamp, assigned doctor clinic room, visit category (General, Maternal, Chronic Follow-up, Acute).
- **What Happens Next:** The token enters the facility live queue.
- **What the Next User Sees:** The triage nurse workstation displays the token in bold with an elapsed wait timer.
- **Why it Matters:** Establishes transparent, orderly patient movement, eliminates hallway congestion, and creates an audit-ready timeline of patient wait durations.
- **Management Monitoring:** Monitors average check-in speed and total queue volume across morning and evening sessions.

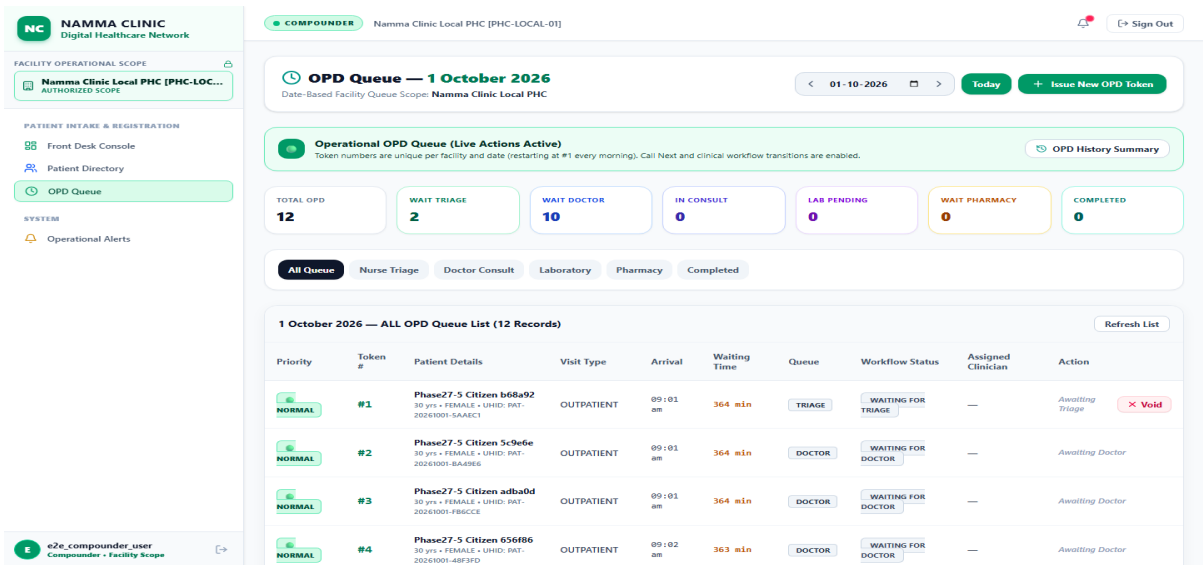


Figure 28 - Facility OPD Token Queue View - Live queue displaying sequential tokens, tracking citizen progression through registration, triage, consultation, laboratory testing, and pharmacy dispensing.

Stage 3: Nurse Triage & Vital Signs Measurement

- **Who Performs It:** Staff Nurse.
- **What They Do:** Calls the patient into the triage room, conducts standardized physiological measurements, enters vital signs, and assesses clinical urgency.
- **Information Captured:** Systolic and diastolic Blood Pressure (mmHg), Pulse Rate (bpm), Body Temperature (°F), Respiratory Rate, Oxygen Saturation (SpO2 %), Height, Weight, and calculated Body Mass Index (BMI). Urgency classification (Green = Routine, Amber = Urgent, Red = Emergency).
- **What Happens Next:** The system automatically computes BMI and flags out-of-range values. The patient moves to the Doctor's consultation waiting queue.
- **What the Next User Sees:** The Doctor sees the token immediately on their queue, highlighted with the nurse's acuity color badge and vital signs summary.
- **Why it Matters:** Ensures unstable patients (e.g., severe hypertension, high fever, hypoxia) are immediately prioritized rather than waiting behind routine mild cases.
- **Management Monitoring:** Reviews vital signs capture compliance rates and identifies triage acuity distributions across clinics.

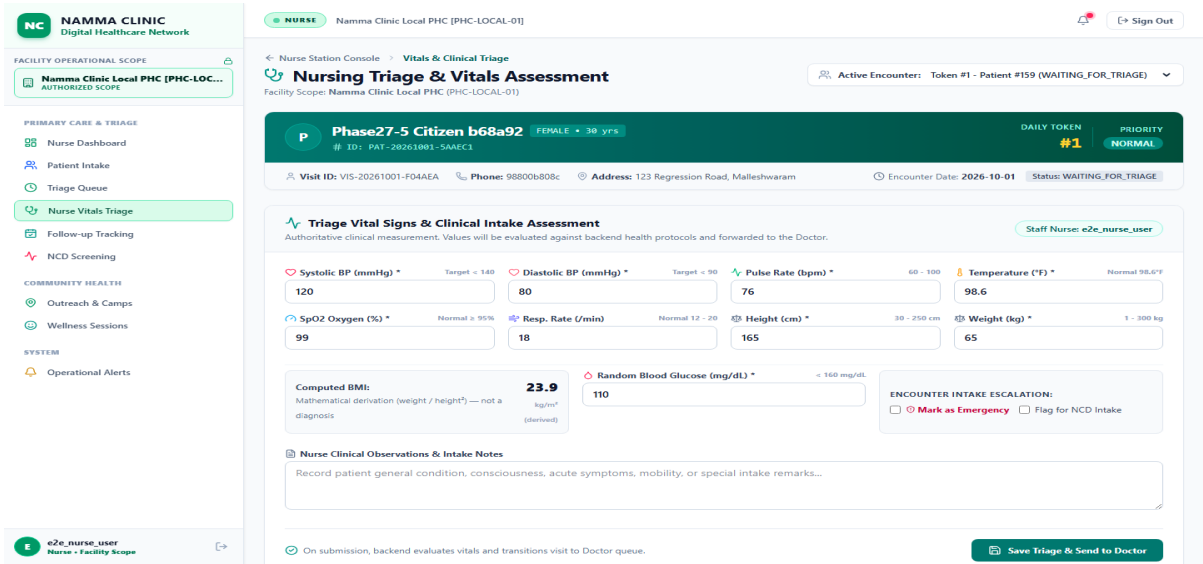


Figure 29 - Nurse Triage Vitals Documentation Screen - Standardized nursing triage form recording blood pressure, pulse, temperature, SpO2, and BMI, with automated clinical acuity color flags.

Stage 4: Doctor Consultation & Clinical Diagnosis

- **Who Performs It:** Medical Officer (Doctor).
- **What They Do:** Calls the patient into the consultation room, reviews nurse-entered vitals, records symptoms and clinical history, conducts physical examination, enters formal diagnoses, orders diagnostic lab tests, and creates electronic prescriptions.
- **Information Captured:** Chief complaints, duration of illness, clinical examination findings, standardized diagnosis codes, lab investigation requisitions, prescribed medications (drug name, strength, dosage form, frequency, duration, food relation, and quantity), and follow-up revisit dates.
- **What Happens Next:** Lab orders instantly transmit to the diagnostic laboratory. Prescriptions transmit electronically to the clinic pharmacy.
- **What the Next User Sees:** The Lab Technician sees ordered tests in their pending queue; the Pharmacist sees the electronic prescription waiting for dispensing.
- **Why it Matters:** Completely replaces illegible paper slips, enforces standardized clinical diagnostic terminology, and automatically prevents drug dosage and frequency misunderstandings.
- **Management Monitoring:** Evaluates doctor consultation throughput, diagnosis prevalence patterns, and antibiotic prescribing compliance.

The screenshot displays the 'Medical Officer Consultation Screen' within the Namma Clinic Digital Healthcare Platform. The interface is divided into a sidebar and a main content area. The sidebar on the left contains navigation links for 'FACILITY OPERATIONAL SCOPE', 'Namma Clinic Local PHC [PHC-LOC...]', 'CLINICAL CARE' (including Doctor Dashboard, Patients, Consultation Queue, Doctor Consultation, Diagnostic Orders, Referral Network, Follow-up Care, NCD Longitudinal Care), and 'SYSTEM' (Clinical Alerts). The main content area is titled 'Doctor Dashboard' and 'Clinical Encounter & Consultation'. It shows the 'Active Encounter' as 'Token #1 - Phase27-5 Citizen b68a92 (WAITING_FOR_TRIAGE)'. Patient details for 'Phase27-5 Citizen b68a92' are listed, including Patient ID, Visit ID, Mobile, and Priority. Below this, there's a section for 'Pre-Consultation Nursing Triage Vitals' with a note that vitals have not been recorded yet. The 'Patient Longitudinal History (15 previous records)' section lists three visits: Visit #155 (Fever, unspecified / Acute viral syndrome), Visit #158 (Type 2 Diabetes Mellitus / Essential Hypertension), and Visit #157 (Tension Headache). The 'Medical Officer Consultation Documentation' section includes 'Chief Complaint' and 'History of Present Illness & Comorbidities' with text input fields. At the bottom, there's a section for 'Clinical Assessment & Physical Examination Findings' with a text input field.

Figure 30 - Medical Officer Consultation Screen - Integrated clinician screen combining chief complaints, past visits, physical examination, standardized diagnoses, and electronic prescription generation.

Stage 5: Diagnostic Laboratory Processing

- **Who Performs It:** Diagnostic Lab Technician.
- **What They Do:** Collects biological specimens (blood, urine, sputum), verifies specimen labeling against patient token, executes diagnostic tests, and enters quantitative results.
- **Information Captured:** Specimen collection timestamp, test result values (e.g., Random Blood Sugar: 178 mg/dL, Urine Albumin: 1+), reference range comparisons, abnormal value alerts, and technician sign-off.
- **What Happens Next:** Results are saved and instantly reflected in the Doctor's consultation chart. The patient returns to the doctor or proceeds to pharmacy.
- **What the Next User Sees:** The Doctor receives a notification that lab results are ready for clinical interpretation and final prescription adjustment.
- **Why it Matters:** Eliminates lost paper lab slips, provides rapid turnaround for point-of-care testing, and flags critical values immediately.
- **Management Monitoring:** Tracks lab test turnaround times, test volume by category, and reagent consumption patterns.

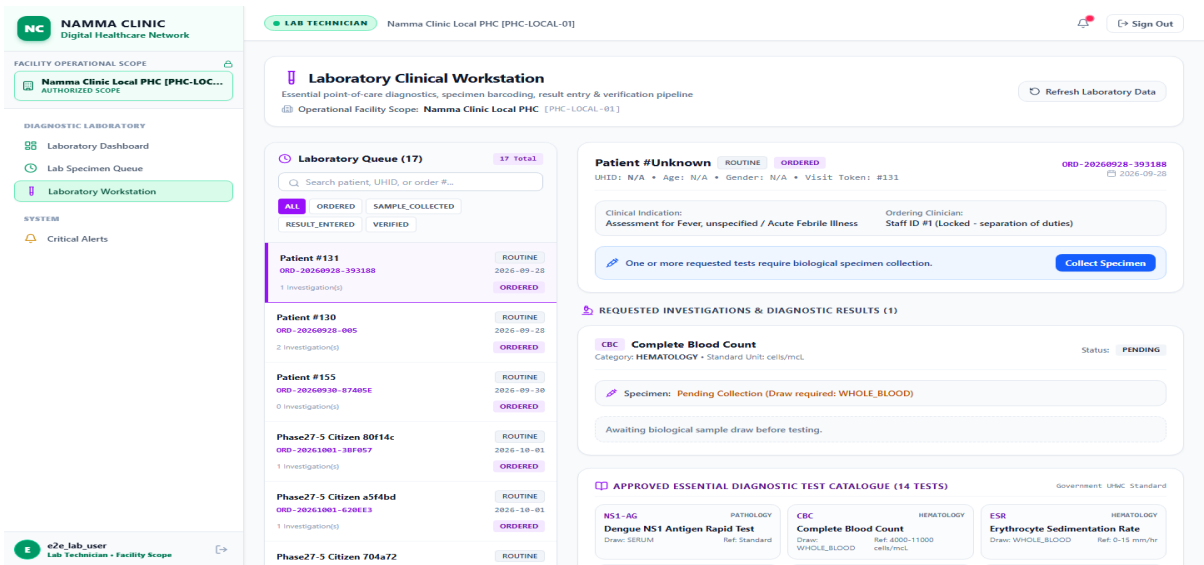


Figure 31 - Diagnostic Test Order & Specimen Intake Screen - Laboratory interface for specimen intake, test execution, quantitative result recording, and instant clinician alert delivery.

Stage 6: Prescription Verification & FEFO Dispensing

- **Who Performs It:** Pharmacist (or Pharmacist/Inventory Officer under dual-role rule).
- **What They Do:** Reviews doctor electronic prescription on the pharmacy workstation, verifies dosage and directions, inspects system-recommended earliest-expiring active batches, issues medicines, and provides counseling to the patient.
- **Information Captured:** Dispensed batch numbers, expiration dates, unit quantities issued, pharmacist verification timestamp, and counseling confirmation.
- **What Happens Next:** The inventory ledger immediately deducts dispensed quantities from the facility stock balance. The prescription status changes to 'Dispensed'.
- **What the Next User Sees:** The Doctor and Administrator see the prescription marked completed in the longitudinal visit timeline.
- **Why it Matters:** Prevents expired drugs from reaching patients, automates stock deduction, eliminates manual ledger posting, and ensures patients receive accurate dosage instructions.
- **Management Monitoring:** Monitors stock turnover, expired drug avoidance rates, and daily pharmaceutical dispensing counts.

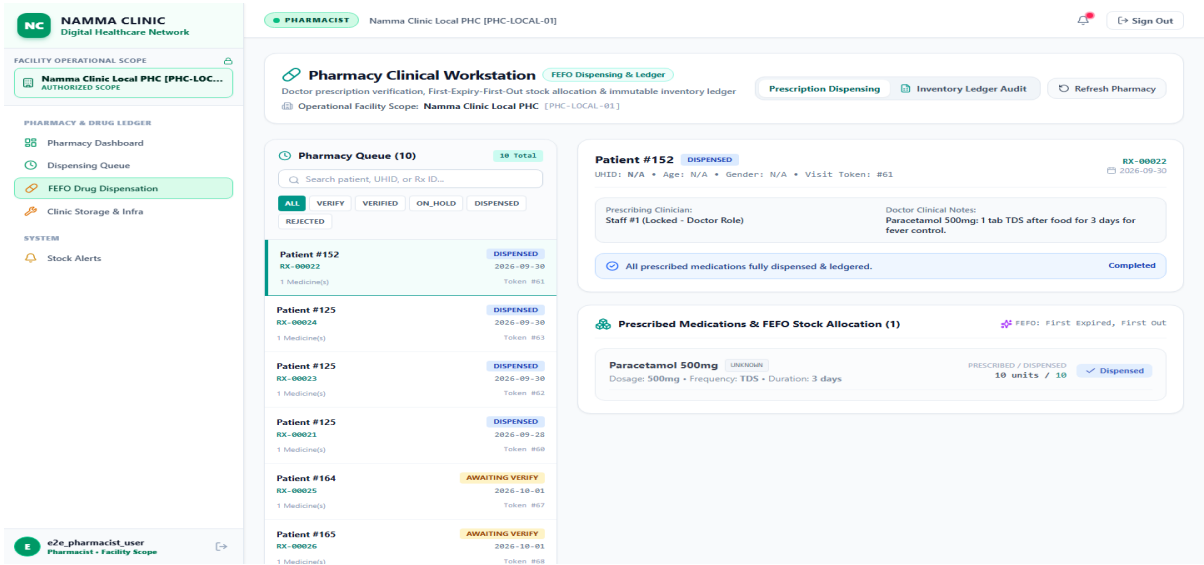


Figure 32 - Pharmacy Prescription Verification & Dispensing Screen - Pharmacist validation screen displaying doctor electronic prescriptions, system-recommended FEFO batches, and medicine counseling notes.

Stage 7: Visit Completion & Longitudinal Record

- **Who Performs It:** System-wide automated consolidation.

- **What They Do:** Consolidates registration demographics, nursing vitals, doctor notes, lab findings, and dispensed drugs into a permanent, longitudinal electronic health timeline.
- **Information Captured:** Total visit duration, complete clinical care summary, and scheduled follow-up reminders.
- **What Happens Next:** If a follow-up revisit is scheduled, the patient is entered into the nurse's chronic care recall register.
- **What the Next User Sees:** At the patient's next visit (even months later), any clinician across the clinic immediately views the complete chronological history.
- **Why it Matters:** Ensures continuous, coordinated healthcare rather than fragmented single encounters. Enables long-term chronic disease management.
- **Management Monitoring:** Computes facility overall turnaround time, patient satisfaction indicators, and clinic completion rates.

PART E - CLINIC OPERATIONS REFERENCE

This section provides a structured operational reference for all twenty-one operational modules supporting clinic workflows.

TWENTY-ONE CLINIC MODULES			
PATIENT CARE:	1. Demographic Registry	2. OPD Token & Queue	3. Clinical Triage
	4. Doctor Consultation	5. Diagnostic Lab	6. Pharmacy Dispensing
SUPPLY CHAIN:	7. Warehouse Inventory	8. Procurement & POs	9. Goods Receipt Notes (GRN)
CARE EXTENSIONS:	10. Staff Administration	11. Specialist Referrals	12. Patient Follow-ups
	13. NCD Chronic Registry	14. Public Surveillance	15. Community Outreach Camps
	16. Wellness & Education	17. Quality Indicators	18. Infrastructure & Cold-Chain
GOVERNANCE:	19. Reporting Engine	20. Early Warning Alerts	21. Security & Audit Logging

1. Patient Demographic Registry

- **Purpose:** Maintains the single authoritative record of all registered citizens.
- **Primary Users:** Front Desk Officer, Staff Nurse, Doctor, Administrator.
- **Workflow:** Search by mobile/name -> Review demographic summary -> Update address/contact if changed.
- **Outcome:** Accurate patient demographic records without duplicate files.
- **Client Value:** Faster check-in times and complete demographic visibility across clinic departments.

The screenshot displays the 'Patient Master Directory & Longitudinal EMR Registry' interface. The sidebar on the left contains navigation links: 'Namma Clinic Local PHC [PHC-LOC-01]', 'Patient Intake & Registration', 'Front Desk Console', 'Patient Directory' (highlighted), 'OPD Queue', and 'Operational Alerts'. The main content area features a search bar with the placeholder 'Search by Patient ID, Name, Mobile Number, or ABHA ID...'. Below the search bar is a table with the following columns: Patient ID, Full Name, Demographics, Mobile, ABHA ID (Demo), Registered Facility, Vulnerability Flag, and Actions. The table lists several patient records, each with an 'Issue Token' button in the Actions column.

Patient ID	Full Name	Demographics	Mobile	ABHA ID (Demo)	Registered Facility	Vulnerability Flag	Actions
PAT-20261001-5AAEC1	Phase27-5 Citizen b68a92	30 yrs • FEMALE	988000b89c	N/A	Namma Clinic Local PHC	Slum Resident / Low Income Group	Issue Token
PAT-20261001-BA49E6	Phase27-5 Citizen 5c9ede	30 yrs • FEMALE	9880016591	N/A	Namma Clinic Local PHC	Slum Resident / Low Income Group	Issue Token
PAT-20261001-FB6CCE	Phase27-5 Citizen adba9d	30 yrs • FEMALE	988000af879	N/A	Namma Clinic Local PHC	Slum Resident / Low Income Group	Issue Token
PAT-20261001-48F3FD	Phase27-5 Citizen 656f86	30 yrs • FEMALE	98800da5ca	N/A	Namma Clinic Local PHC	Slum Resident / Low Income Group	Issue Token
PAT-20261001-443BA0	Phase27-5 Citizen 42c987	30 yrs • FEMALE	988003d8bd	N/A	Namma Clinic Local PHC	Slum Resident / Low Income Group	Issue Token
PAT-20261001-5CD289	Phase27-5 Citizen 7f25dc	30 yrs • FEMALE	9880074f23	N/A	Namma Clinic Local PHC	Slum Resident / Low Income Group	Issue Token
PAT-20261001-EC4568	Phase27-5 Citizen 5d51cf	30 yrs • FEMALE	988002d99a	N/A	Namma Clinic Local PHC	Slum Resident / Low Income Group	Issue Token
PAT-20261001-F5A48C	Phase27-5 Citizen 3184e5	30 yrs • FEMALE	98800f1601	N/A	Namma Clinic Local PHC	Slum Resident / Low Income Group	Issue Token

Figure 33 - Patient Demographic Registry Screen - Searchable central directory of registered citizens providing contact verification, demographic updates, and duplicate record prevention across clinic visits.

2. OPD Token & Queue Module

- **Purpose:** Coordinates fair, sequential patient movement between clinic departments.

- **Primary Users:** Front Desk Officer, Nurse, Doctor, Pharmacist.
- **Workflow:** Token issued at reception -> Displayed on waiting queue -> Called into service rooms sequentially.
- **Outcome:** Orderly patient flow with real-time queue visibility.
- **Client Value:** Eliminates waiting room disputes and provides transparent clinic wait times.

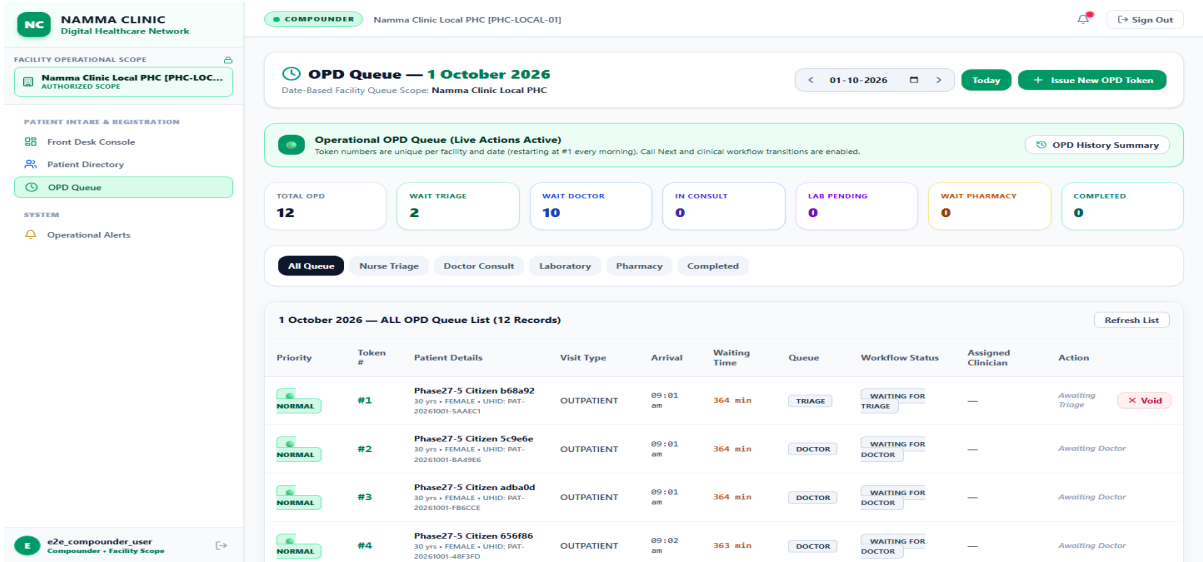


Figure 34 - OPD Token & Waiting Queue Console - Real-time facility waiting monitor organizing citizen flow sequentially across triage, consultation, diagnostic sampling, and pharmacy dispensing.

3. Clinical Triage Module

- **Purpose:** Gathers physiological baseline measurements and identifies urgent walk-in cases.
- **Primary Users:** Staff Nurse.
- **Workflow:** Call token -> Measure vitals -> Calculate BMI -> Assign acuity score -> Route to Doctor.
- **Outcome:** Standardized vitals documentation and rapid prioritization of critical patients.
- **Client Value:** Protects patient safety by preventing critical emergencies from waiting unnoticed in general queues.

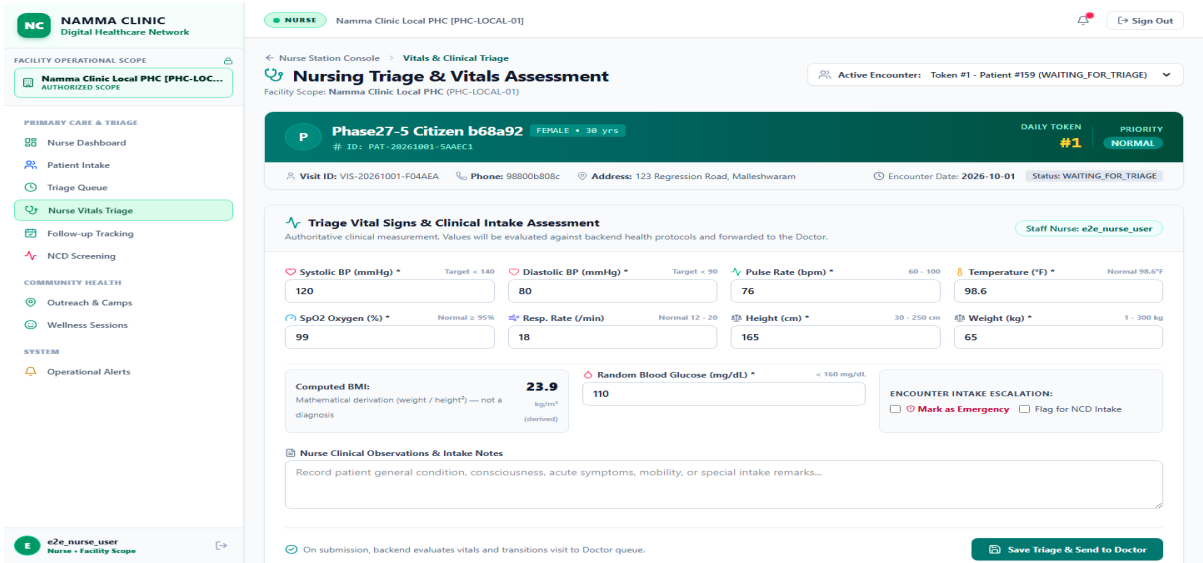


Figure 35 - Clinical Vitals Assessment Form - Nursing documentation tool capturing physiological baselines, calculating BMI, and triggering emergency priority flags for the doctor consultation queue.

4. Doctor Consultation Module

- **Purpose:** Records clinical encounters, physical examinations, diagnoses, and medical orders.
- **Primary Users:** Medical Officer (Doctor).

- **Workflow:** Review vitals -> Record history & examination -> Select diagnosis -> Order labs & prescribe drugs.
- **Outcome:** Standardized, legible clinical consultation record with digital orders.
- **Client Value:** Improves diagnostic consistency, eliminates handwriting errors, and structures clinical data.

The screenshot displays the 'Doctor Clinical Examination & Prescribing Interface'. The top navigation bar includes 'Doctor Dashboard' and 'Clinical Encounter & Consultation'. The active encounter is 'Token #1 - Phase27-5 Citizen b68a92 (WAITING_FOR_TRIAGE)'. The patient information section shows 'Phase27-5 Citizen b68a92', Female, 30 years old, with Patient ID: PAT-20261001-SAAEC1, Visit ID: VIS-20261001-F04EA, Mobile: 98900b808c, and Priority: NORMAL. The 'Pre-Consultation Nursing Triage Vitals' section indicates that triage vitals have not been recorded yet. The 'Patient Longitudinal History (15 previous records)' section lists three visits: Visit #155 (Fever, unspecified / Acute viral syndrome), Visit #158 (Type 2 Diabetes Mellitus / Essential Hypertension), and Visit #157 (Tension Headache). The 'Medical Officer Consultation Documentation' section includes a 'Chief Complaint' (e.g., High fever with chills, body aches for 3 days) and a 'History of Present Illness & Comorbidities' (e.g., Known hypertensive, no history of cough or rash). The 'Clinical Assessment & Physical Examination Findings' section shows 'e.g. Mild pharyngeal congestion, chest clear, no hepatosplenomegaly'.

Figure 36 - Doctor Clinical Examination & Prescribing Interface - Comprehensive consultation workspace recording symptoms, physical findings, ICD-standardized diagnoses, lab orders, and digital prescriptions.

5. Laboratory Module

- **Purpose:** Processes diagnostic investigation orders and logs quantitative test results.
- **Primary Users:** Diagnostic Lab Technician.
- **Workflow:** Receive order -> Collect specimen -> Perform test -> Enter quantitative findings -> Notify doctor.
- **Outcome:** Timely diagnostic results linked directly to the patient's visit.
- **Client Value:** Faster diagnosis confirmation and complete traceability of test requisitions.

The screenshot displays the 'Laboratory Clinical Workstation'. The top navigation bar includes 'Laboratory Dashboard' and 'Laboratory Workstation'. The active encounter is 'Token #1 - Phase27-5 Citizen b68a92 (WAITING_FOR_TRIAGE)'. The patient information section shows 'Phase27-5 Citizen b68a92', Female, 30 years old, with Patient ID: PAT-20261001-SAAEC1, Visit ID: VIS-20261001-F04EA, Mobile: 98900b808c, and Priority: NORMAL. The 'Laboratory Queue (17)' section shows a list of investigations with columns for Patient, ORD, Routine, and Ordered. The 'REQUESTED INVESTIGATIONS & DIAGNOSTIC RESULTS (1)' section shows a 'CBC Complete Blood Count' with a status of 'PENDING'. The 'APPROVED ESSENTIAL DIAGNOSTIC TEST CATALOGUE (14 TESTS)' section lists various tests including NS1-Ag, Dengue NS1 Antigen Rapid Test, CBC Complete Blood Count, and ESR Erythrocyte Sedimentation Rate.

Figure 37 - Diagnostic Laboratory Queue & Results Entry Screen - Specimen intake list enabling rapid result documentation, automatic abnormal range flagging, and instant electronic notification to the treating physician.

6. Pharmacy Dispensing Module

- **Purpose:** Dispenses prescribed medicines safely using First-Expired, First-Out (FEFO) allocation.
- **Primary Users:** Pharmacist.
- **Workflow:** Review doctor prescription -> Confirm FEFO batch allocations -> Dispense stock -> Counsel patient.

- **Outcome:** Accurate drug fulfillment with automated inventory decrement.
- **Client Value:** Eliminates dispensing errors, prevents expired drug usage, and provides complete dispensing records.

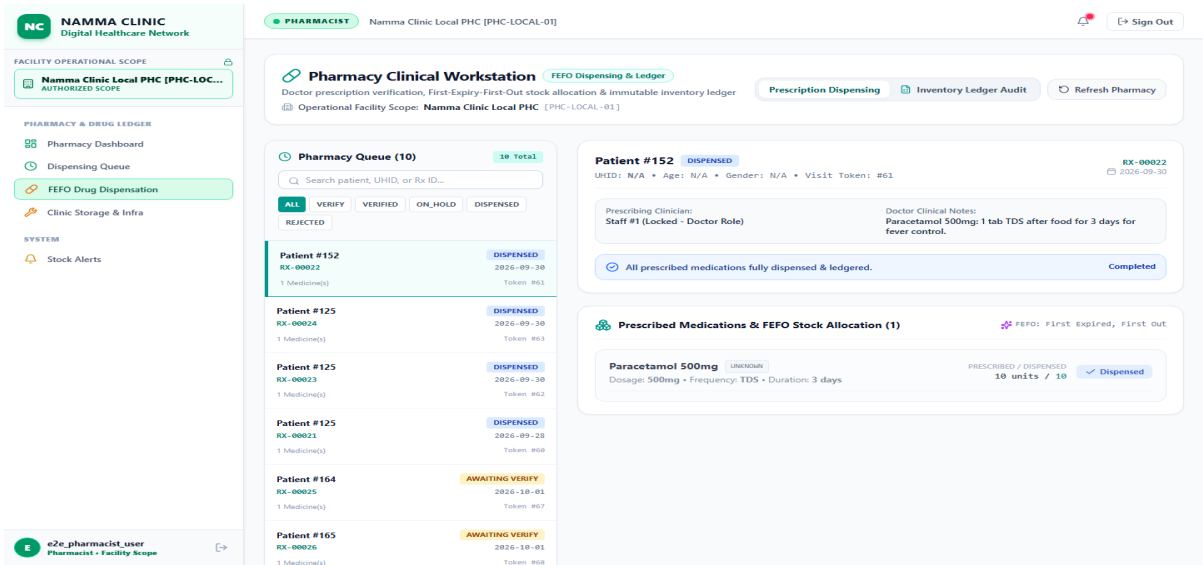


Figure 38 - Pharmacy Outpatient Dispensing Workstation - Dispensing interface matching electronic doctor prescriptions against warehouse inventory with automated First-Expired, First-Out (FEFO) batch allocation.

7. Warehouse Inventory Module

- **Purpose:** Maintains real-time facility stock ledgers, batch numbers, and reorder levels.
- **Primary Users:** Inventory Officer, Pharmacist.
- **Workflow:** Monitor balances -> Inspect reorder warnings -> Audit physical counts against system balances.
- **Outcome:** Accurate stock levels with complete batch-level traceability.
- **Client Value:** Prevents unexpected stockouts of vital life-saving drugs and eliminates expired medicine waste.

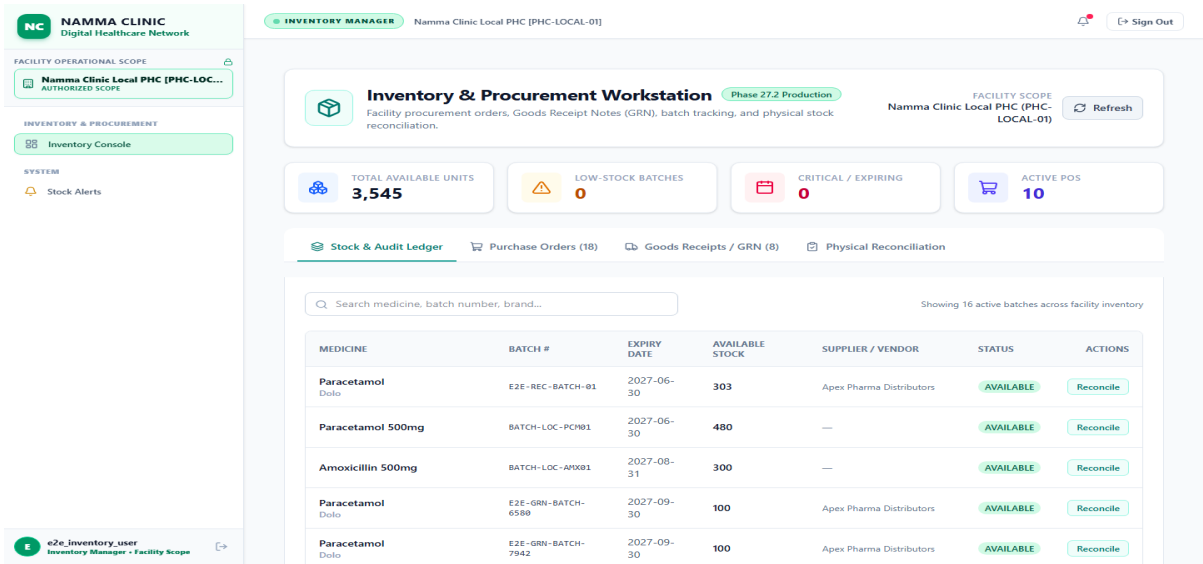


Figure 39 - Clinic Warehouse Stock Ledger - Comprehensive facility store ledger displaying active stock on hand, reorder thresholds, supplier batch tracking, and automated expiration warnings.

8. Procurement & Purchase Orders

- **Purpose:** Generates and tracks supplier purchase orders for medicines and clinical consumables.
- **Primary Users:** Inventory Officer, Administrator.
- **Workflow:** Review depleted stock -> Generate PO with quantities and approved suppliers -> Track vendor delivery.
- **Outcome:** Structured procurement cycle with vendor delivery tracking.

- **Client Value:** Transparent procurement timelines and orderly replenishment cycles.

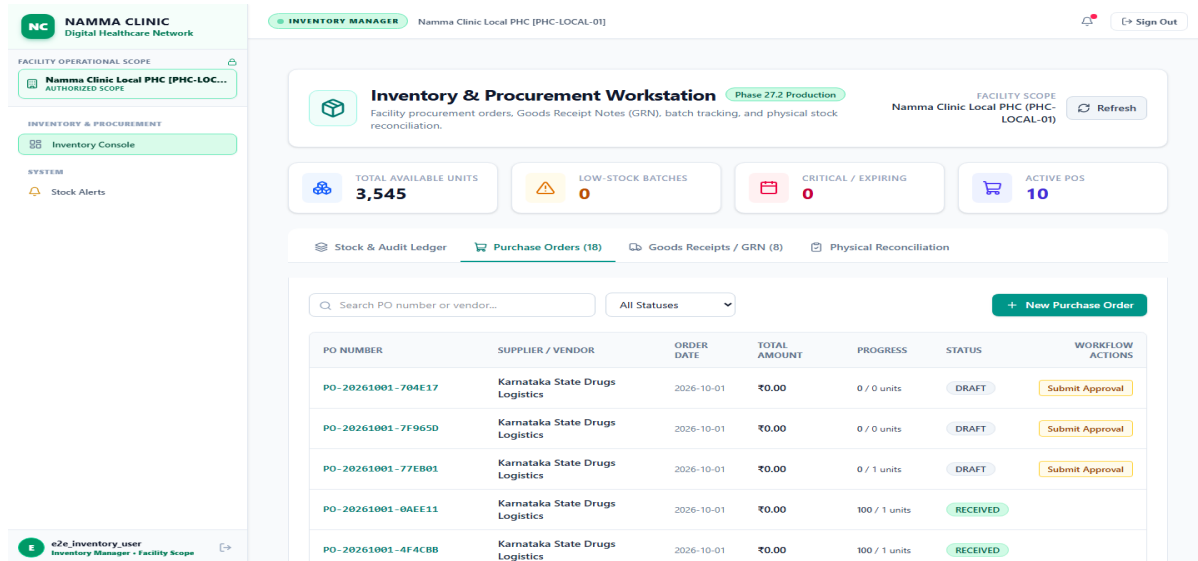


Figure 40 - Procurement Requisition & Purchase Order Tracker - Supplier purchasing console tracking stock replenishments, order quantities, delivery commitments, and vendor fulfillment status.

9. Goods Receipt Notes (GRN) Module

- **Purpose:** Formally receives incoming vendor shipments and logs batches into clinic inventory.
- **Primary Users:** Inventory Officer.
- **Workflow:** Inspect delivery package -> Verify batch numbers, quantities & expiry dates -> Issue GRN receipt.
- **Outcome:** Incoming stock immediately active in inventory with verified expiry dates.
- **Client Value:** Eliminates inventory shrinkage upon receipt and ensures uninspected stock is never dispensed.

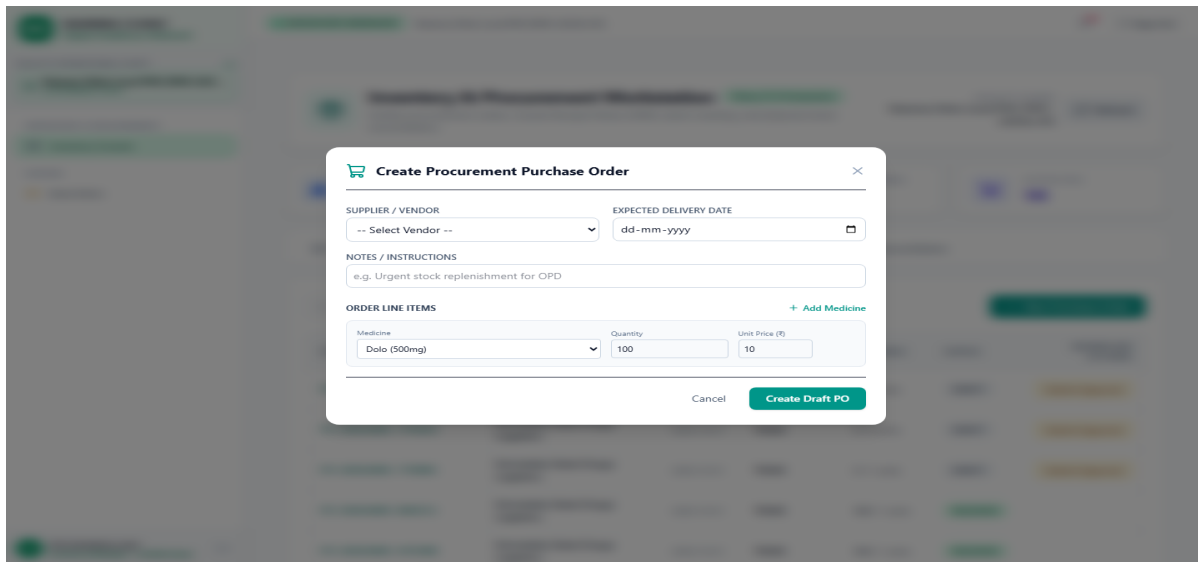


Figure 41 - Goods Receipt Note (GRN) Inward Verification Modal - Formal delivery receipt workflow capturing manufacturer batch identifiers, expiration dates, unit counts, and physical quality verification.

10. Staff Administration Module

- **Purpose:** Manages healthcare personnel accounts, role allocations, and facility scoping.
- **Primary Users:** Clinic Administrator, District Health Officer.
- **Workflow:** Create staff profile -> Assign standard operational roles -> Enable/disable accounts.
- **Outcome:** Verified staff role assignments governed by administrative security.
- **Client Value:** Strict operational governance and audit-compliant workforce management.

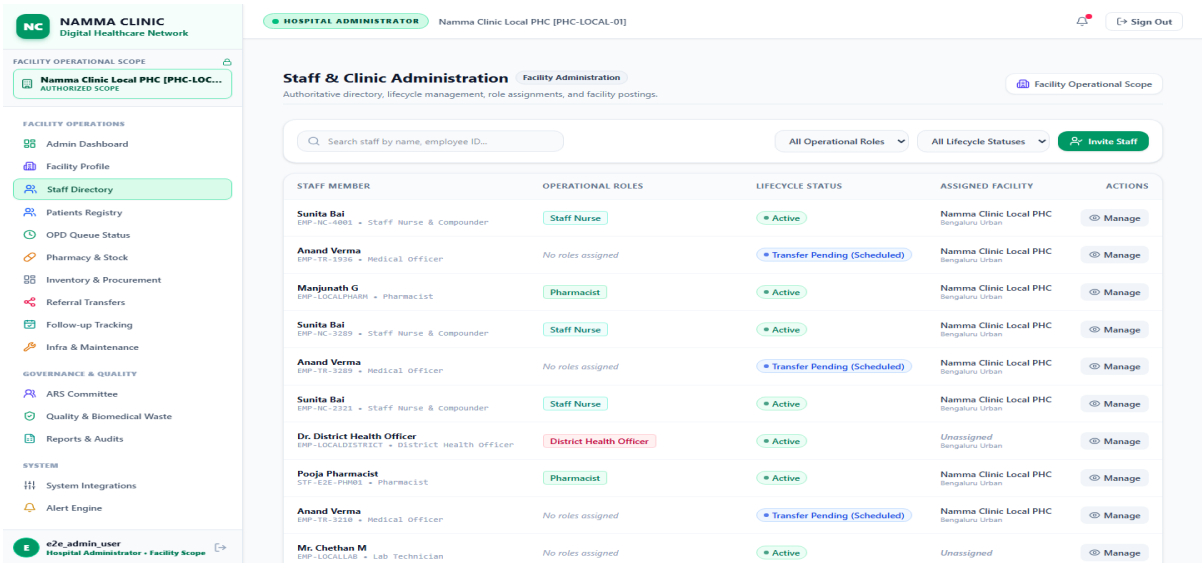


Figure 42 - Facility Staff Management & Role Allocation - Personnel governance screen allowing administrators to configure user profiles, grant verified operational roles, and enforce facility access scoping.

11. Referrals Management Module

- **Purpose:** Coordinates outbound patient transfers to secondary and tertiary hospitals.
- **Primary Users:** Medical Officer, Clinic Administrator.
- **Workflow:** Initiate referral with clinical summary -> Select target hospital -> Monitor transfer progress.
- **Outcome:** Structured referral documentation with clear clinical reason for transfer.
- **Client Value:** Prevents patients from getting lost in the broader hospital network and provides clinical continuity.

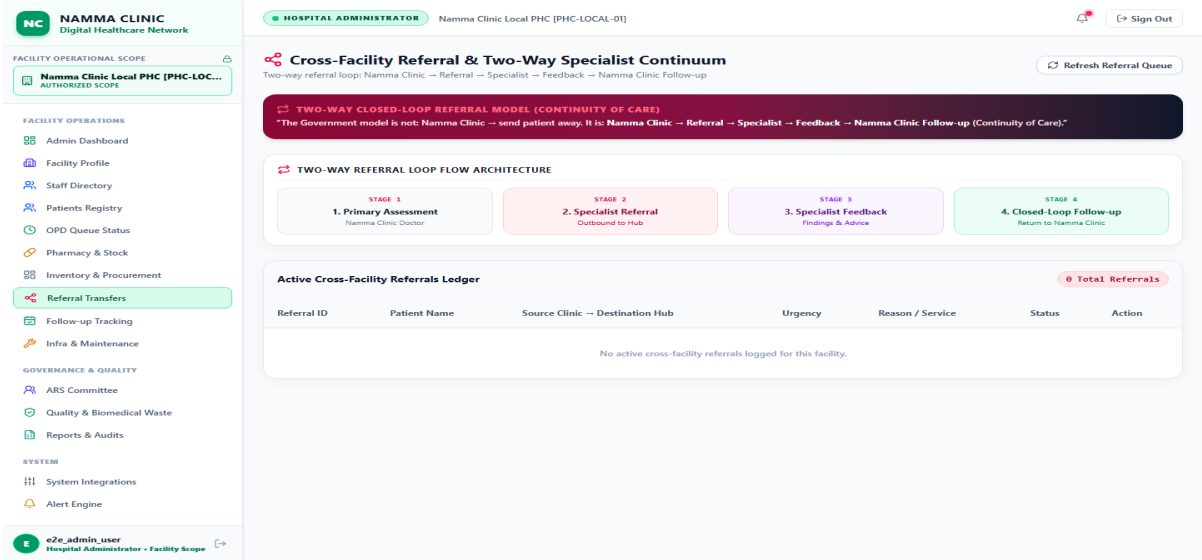


Figure 43 - Outbound Referral Coordination Console - Referral management dashboard facilitating patient transfers to secondary hospitals and tracking tertiary care follow-up outcomes.

12. Patient Follow-ups Module

- **Purpose:** Manages scheduled recall dates for chronic disease patients and post-acute reviews.
- **Primary Users:** Staff Nurse, Medical Officer.
- **Workflow:** View daily due list -> Contact patients due for review -> Mark completed upon arrival.
- **Outcome:** Timely follow-up care for chronic hypertension, diabetes, and post-acute patients.
- **Client Value:** Drastically improves chronic disease control rates and reduces hospital readmissions.

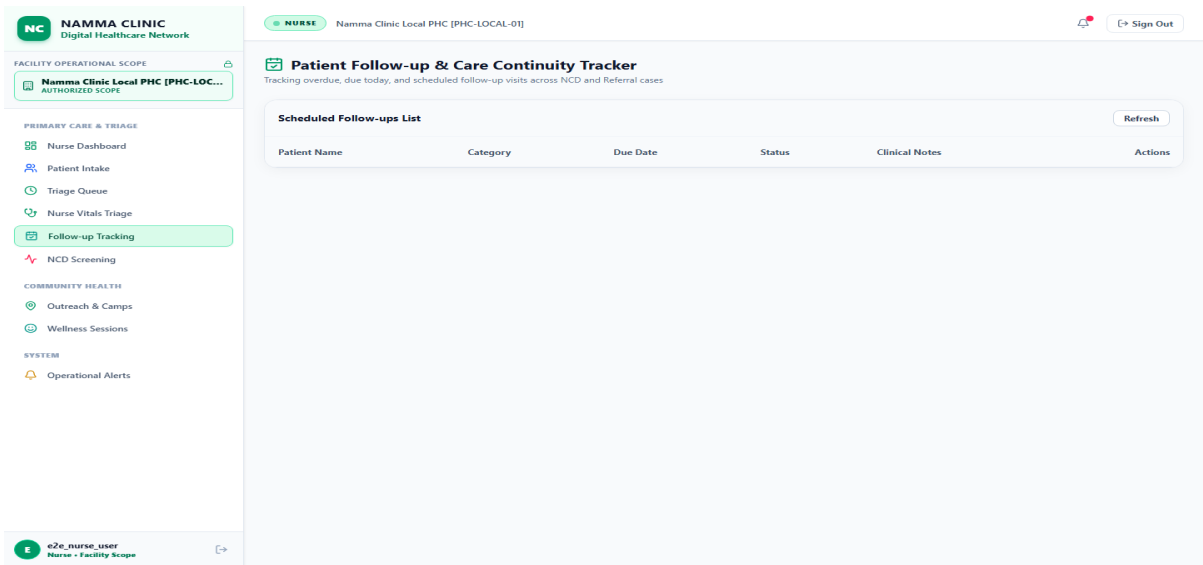


Figure 44 - Patient Follow-up & Recall Workstation - Longitudinal care calendar tracking scheduled revisit dates for hypertensive, diabetic, and post-acute review patients.

13. Non-Communicable Diseases (NCD) Registry

- **Purpose:** Tracks long-term management of hypertensive, diabetic, and chronic care cohorts.
- **Primary Users:** Staff Nurse, Medical Officer.
- **Workflow:** Identify chronic indicators -> Enroll patient in NCD registry -> Track BP and glucose trends over time.
- **Outcome:** Structured chronic disease management with historical control indicators.
- **Client Value:** Enables population-level management of common chronic conditions.

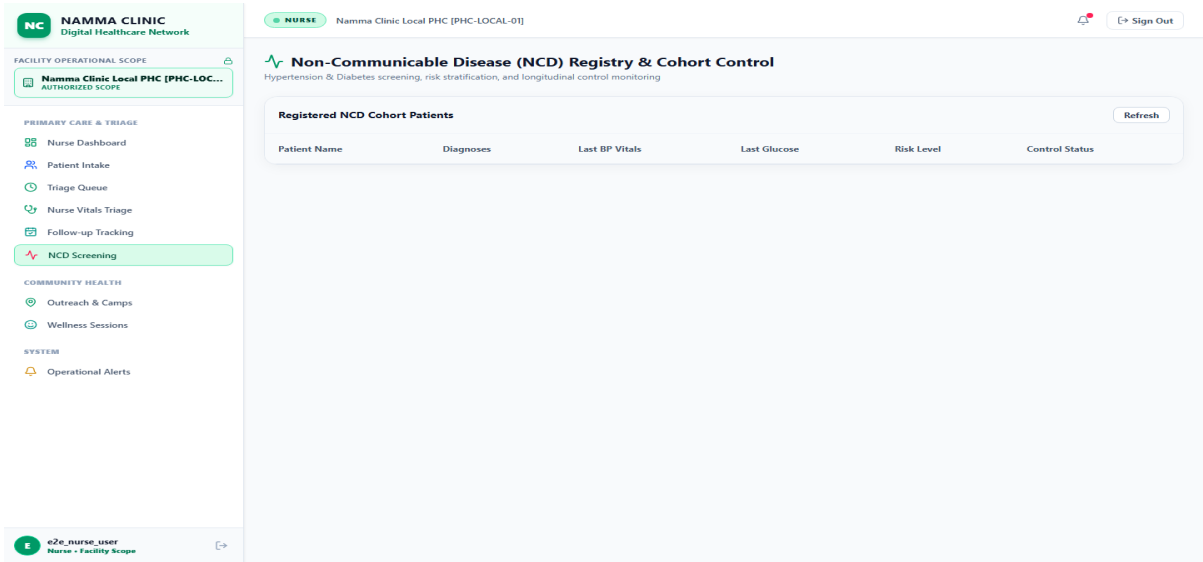


Figure 45 - Non-Communicable Disease (NCD) Registry Screen - Chronic patient monitoring registry tracking community screening results, blood pressure trends, and glycemic management over time.

14. Public Health Surveillance Module

- **Purpose:** Monitors disease trends, fever spikes, and communicable outbreak signals.
- **Primary Users:** District Health Officer, Administrator.
- **Workflow:** Diagnoses aggregate in real time -> System flags unusual syndromic clusters -> Health officers dispatch response teams.
- **Outcome:** Early outbreak warning across urban wards and neighborhood clusters.
- **Client Value:** Rapid epidemic containment and proactive public health response.

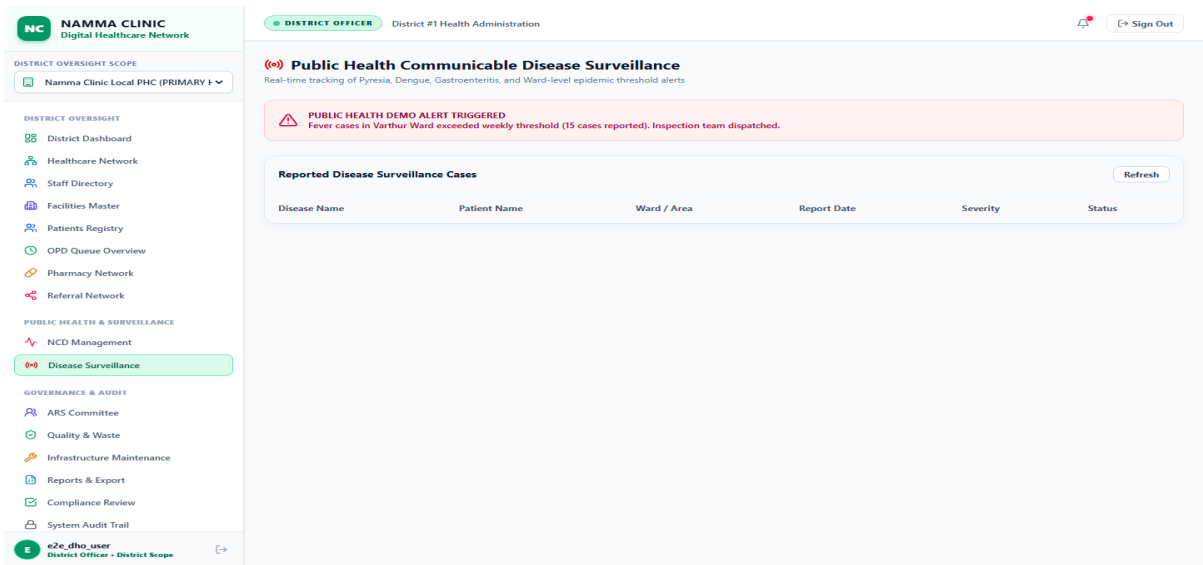


Figure 46 - Public Health Epidemiological Surveillance Console - Automated syndromic surveillance map tracking acute fever, diarrheal illness, and vector-borne clusters across the health network.

15. Community Outreach Module

- **Purpose:** Plans and documents health screening camps in community halls, schools, and workplaces.
- **Primary Users:** Staff Nurse, Medical Officer.
- **Workflow:** Schedule camp -> Record citizen screenings on-site -> Flag identified cases for clinic follow-up.
- **Outcome:** Documented community screening sessions with clear patient follow-up pathways.
- **Client Value:** Extends healthcare access directly into underserved neighborhood communities.

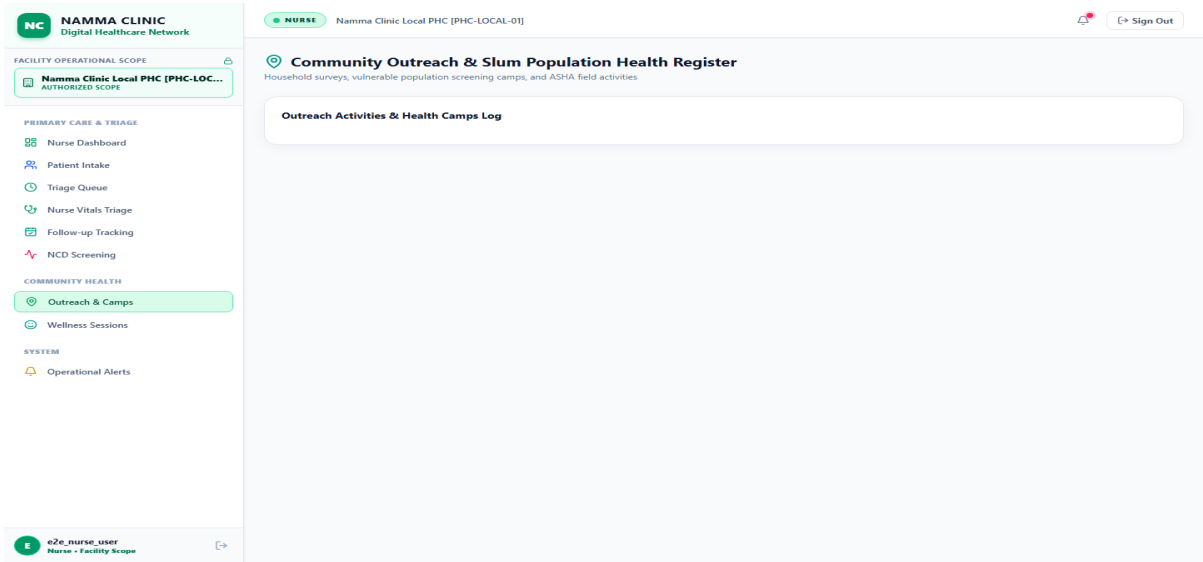


Figure 47 - Community Health Outreach Camp Documentation Screen - Mobile camp tracking interface recording field screening dates, target demographics, and primary health outreach services delivered.

16. Wellness & Health Education Module

- **Purpose:** Logs lifestyle interventions, preventive counseling, yoga, and dietary education.
- **Primary Users:** Staff Nurse, Medical Officer.
- **Workflow:** Record patient counseling session -> Document lifestyle guidance provided -> Track follow-up adherence.
- **Outcome:** Documented preventive health interventions integrated into medical history.
- **Client Value:** Promotes wellness-first healthcare delivery and proactive illness prevention.

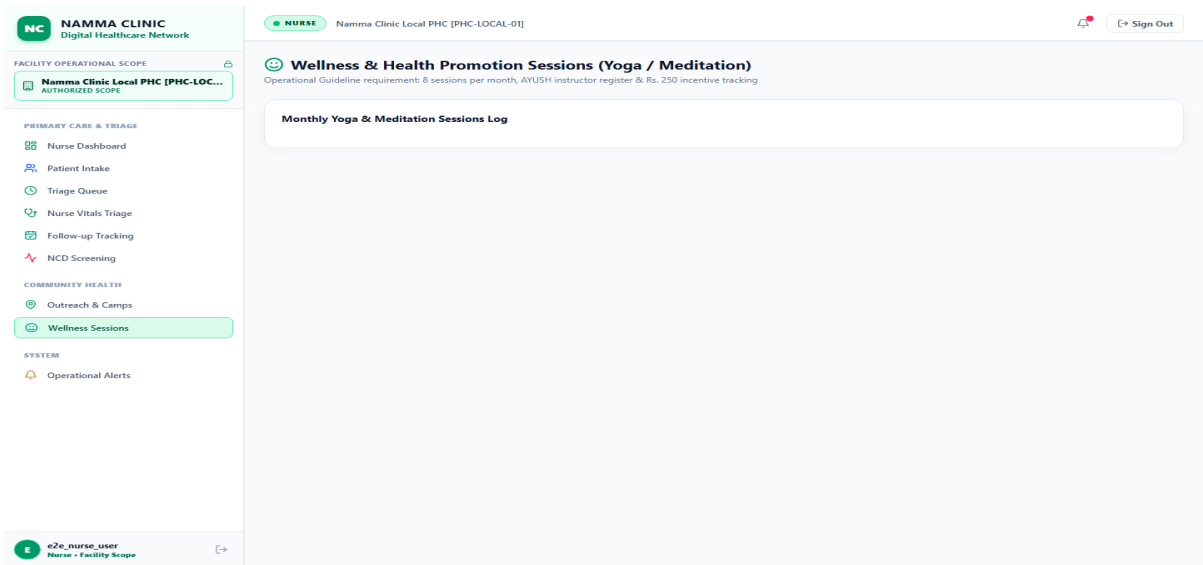


Figure 48 - Preventive Wellness & Health Counseling Screen - Patient education dashboard documenting lifestyle counseling sessions, smoking cessation support, and dietary guidance delivered.

17. Quality Indicators & Facility KPIs

- **Purpose:** Monitors clinical service benchmarks, waiting times, and patient throughput standards.
- **Primary Users:** Clinic Administrator, District Health Officer.
- **Workflow:** Track live KPIs -> Identify bottlenecks in triage or pharmacy -> Implement operational improvements.
- **Outcome:** Objective quality metrics comparing facility performance over time.
- **Client Value:** Continuous healthcare delivery improvement and accountability to service charters.

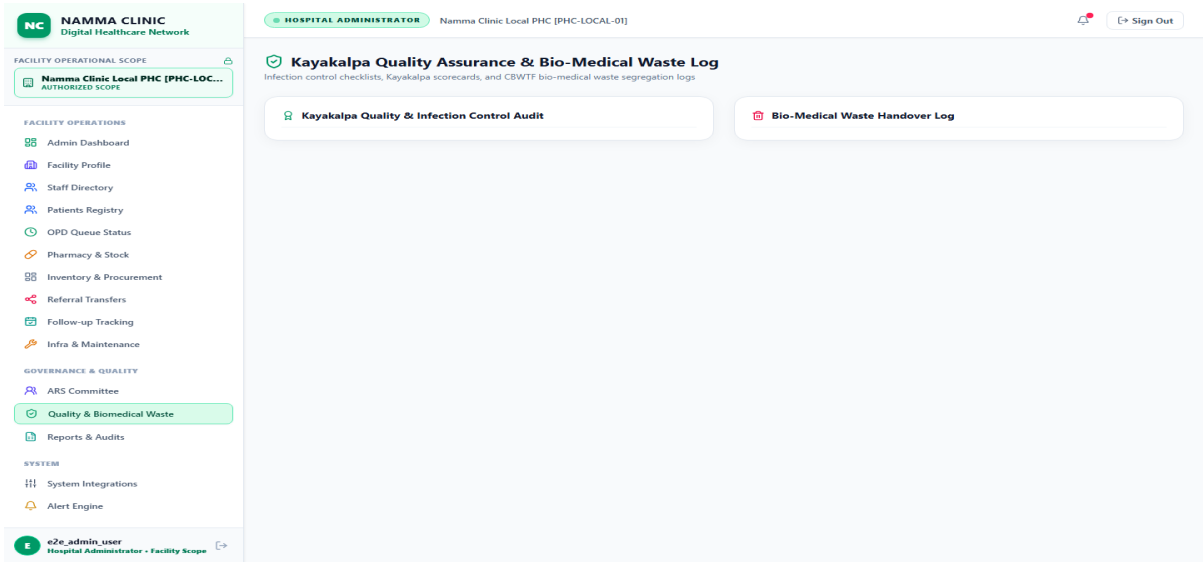


Figure 49 - Clinic Quality Indicators & Performance Dashboard - Facility quality tracking screen showing average consultation durations, patient wait times, prescription completion rates, and service KPIs.

18. Clinic Infrastructure & Storage Module

- **Purpose:** Tracks facility equipment, vaccine cold-chain storage temperatures, and backup power.
- **Primary Users:** Clinic Administrator, Inventory Officer.
- **Workflow:** Log daily refrigerator temperatures -> Record routine equipment servicing -> Flag maintenance needs.
- **Outcome:** Compliant cold-chain storage and operational clinical assets.
- **Client Value:** Protects vaccine potency and guarantees uninterrupted clinic service capability.

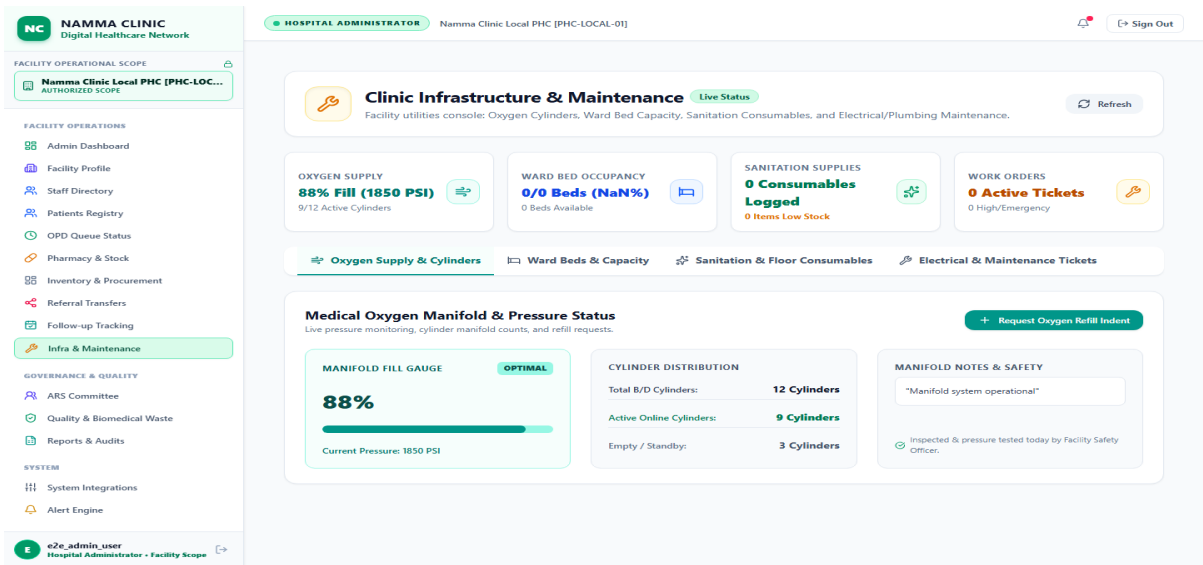


Figure 50 - Infrastructure & Cold-Chain Storage Monitoring - Facility asset registry tracking cold-chain vaccine refrigerators, diagnostic analyzers, backup power, and routine maintenance logs.

19. Operational Reporting Engine

- **Purpose:** Generates daily attendance registers, pharmacy consumption reports, and disease summaries.
- **Primary Users:** Clinic Administrator, District Health Officer.
- **Workflow:** Select reporting period -> Filter by department or service -> Generate and export verified operational reports.
- **Outcome:** Instant administrative summaries without manual register counting.
- **Client Value:** Saves dozens of staff hours per week and eliminates manual reporting discrepancies.

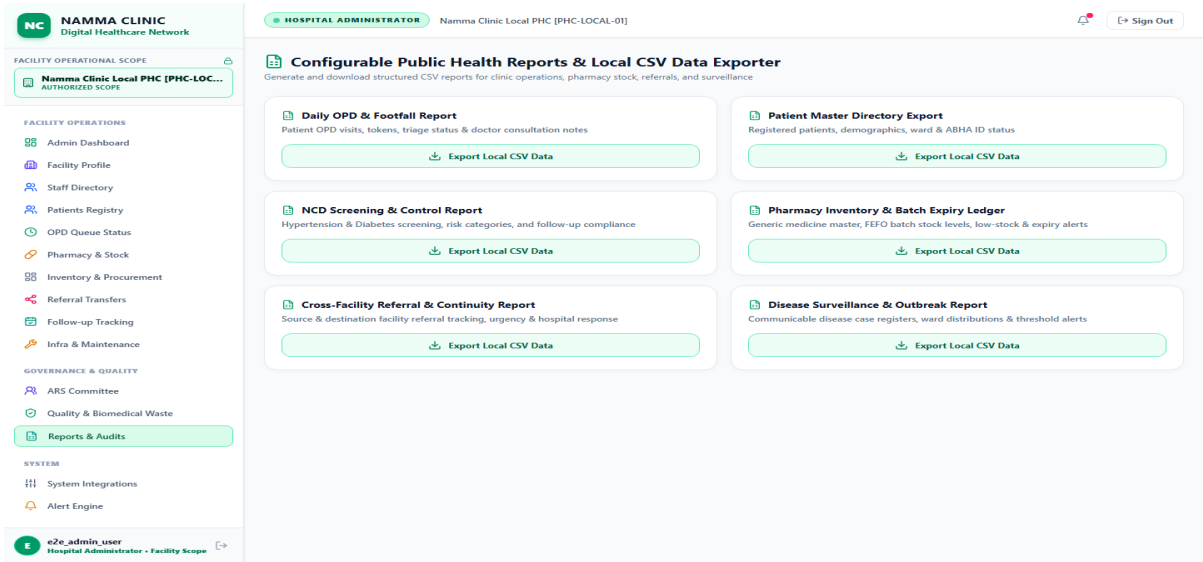


Figure 51 - Operational & Administrative Reporting Engine - Standardized analytics suite generating daily patient attendance tallies, pharmacy stock turnover, and disease incidence reports.

20. Alerts & Early Warning Hub

- **Purpose:** Delivers real-time notifications for critical lab findings, vital signs alerts, and stockout warnings.
- **Primary Users:** All Staff Roles.
- **Workflow:** Critical event occurs -> Alert appears on responsible workstation -> User acknowledges and acts.
- **Outcome:** Immediate awareness of urgent operational and clinical events.
- **Client Value:** Rapid clinical escalation for patient emergencies and zero surprise stockouts.

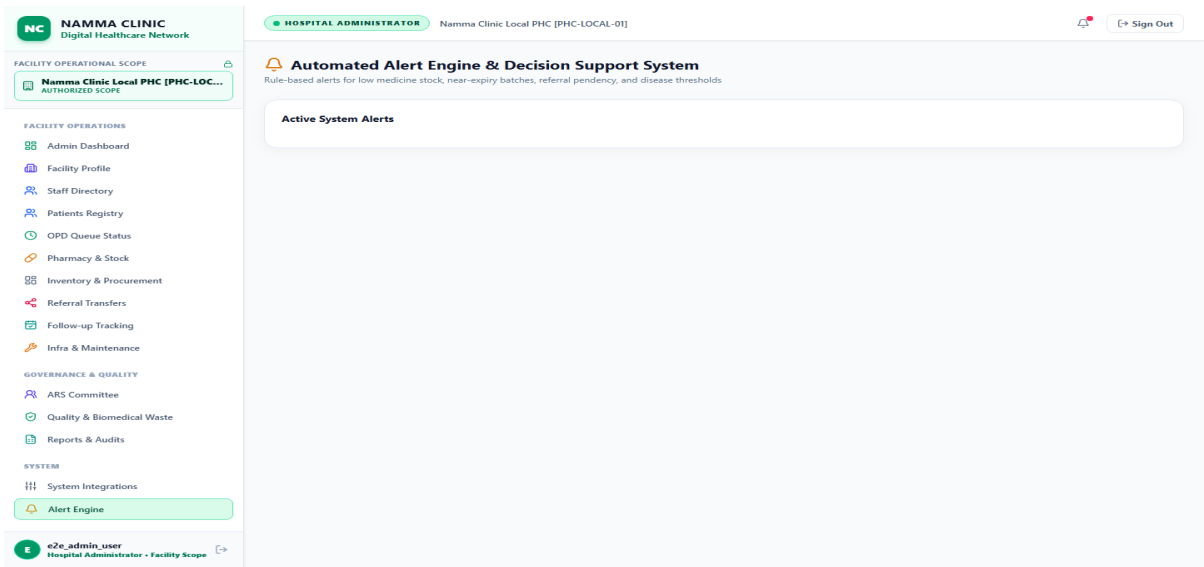


Figure 52 - Real-time Operational Alerts & Notifications - Unified alerting console notifying healthcare teams of critical lab results, urgent triage scores, and low drug inventory thresholds.

21. Security & Audit Logging

- **Purpose:** Maintains an immutable chronological audit trail of all logins, record views, and transactions.
- **Primary Users:** District Health Officer, System Administrator.
- **Workflow:** User performs action -> Event is recorded with timestamp and role -> Auditor reviews logs.
- **Outcome:** Complete, unalterable accountability for every record accessed or modified.
- **Client Value:** Guarantees patient privacy compliance and eliminates administrative denial of actions.

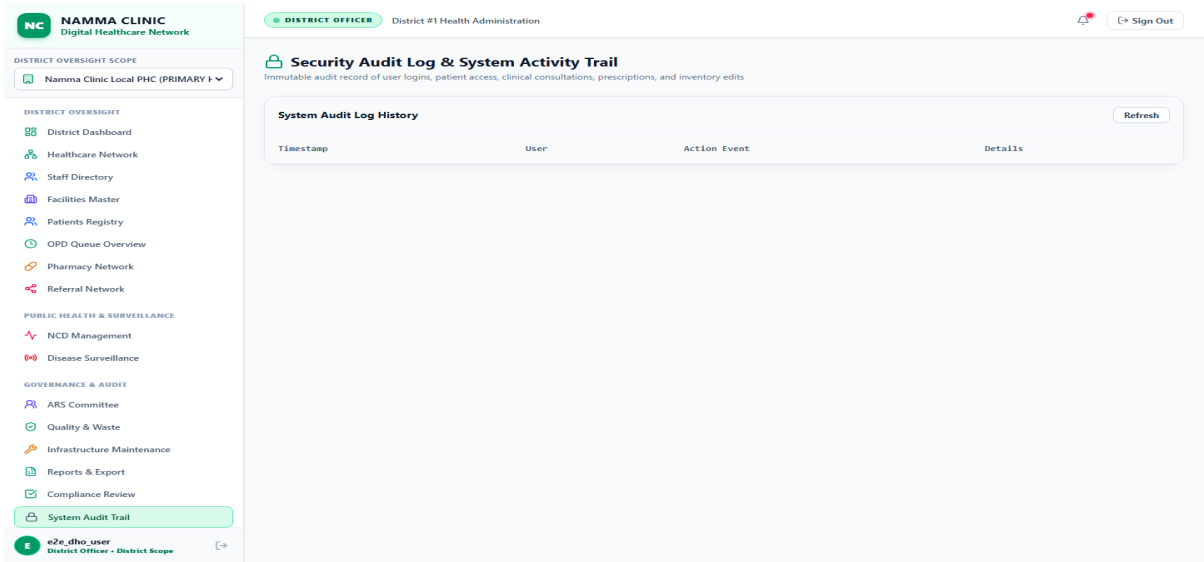
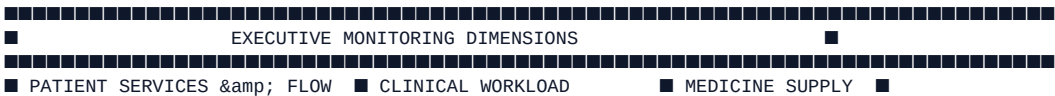


Figure 53 - Security Audit Trail Console - Immutable digital event log documenting user authentication, role assumption, clinical records access, and administrative actions.

PART F - MANAGEMENT & DHO EXECUTIVE VIEW

Health Department Leadership and District Health Officers need clear, actionable visibility into healthcare delivery across their network. Namma Clinic translates frontline clinic data into executive intelligence.



■ • Daily OPD Volume	■ • Consultations per Doc	■ • Warehouse Stock	■
■ • Average Wait Times	■ • Diagnostic Test Volume	■ • Expiry Risk	■
■ • Triage Acuity Ratio	■ • Antibiotic Usage Rate	■ • Stockout Flags	■
■ PUBLIC HEALTH TRENDS	■ WORKFORCE & POSTINGS	■ GOVERNANCE & ARS	■
■ • Disease Outbreak Maps	■ • Doctor Attendance	■ • Fund Balances	■
■ • Chronic NCD Screening	■ • Single-Nurse Staffing	■ • Audit Activity	■
■ • Slum Outreach Coverage	■ • Vacant Clinic Roles	■ • Quality Scores	■

What Management Monitors & What it Tells Leadership

1. Patient Attendance Trends:

- *What the screen shows:* Hourly walk-in arrivals, new citizen registrations, and total visits by clinic.
- *What this tells leadership:* Identifies which clinics are facing high patient loads and where additional doctors or extended operating hours are required.

2. Clinic Waiting Times & Bottlenecks:

- *What the screen shows:* Average minutes from registration to triage, triage to consultation, and consultation to medication collection.
- *What this tells leadership:* Pinpoints exact workflow delays - revealing whether a clinic's bottleneck is at doctor consultation, laboratory testing, or the dispensing counter.

3. Clinical Quality & Antibiotic Stewardship:

- *What the screen shows:* Percentage of patient visits resulting in antibiotic prescriptions, injection rates, and adherence to standard clinical guidelines.
- *What this tells leadership:* Helps health leaders prevent antibiotic overuse and identify clinics requiring clinical training.

4. District Pharmaceutical Stock Health:

- *What the screen shows:* Value and quantity of medicines expiring within 30, 60, and 90 days across all facilities; count of facilities with zero stock of essential drugs.
- *What this tells leadership:* Enables proactive stock rebalancing - moving near-expiry medicines from quiet clinics to busy urban health centers before they expire.

5. Syndromic Epidemiological Outbreak Alerts:

- *What the screen shows:* Spikes in presenting symptoms (fever with chills, acute respiratory distress, acute watery diarrhea) mapped by neighborhood pin code.
- *What this tells leadership:* Provides early warning of vector-borne (Dengue, Malaria) or waterborne (Cholera) disease outbreaks days before hospital admissions surge.

6. Workforce Attendance & Deployment:

- *What the screen shows:* Active staff on duty, facilities operating under dual-role staffing, and vacant medical officer positions.
- *What this tells leadership:* Gives directors real-time visibility into staff attendance without waiting for monthly attendance registers.

PART G - SECURITY, ACCESS CONTROL & RESPONSIBILITY

Information security in Namma Clinic is designed around real healthcare operations, protecting patient privacy while ensuring clinical staff have immediate access to necessary medical data.

ROLE-BASED SECURITY & PRIVACY ENFORCEMENT			
[FRONT DESK OFFICER]	[STAFF NURSE]	[MEDICAL OFFICER]	[PHARMACIST]
• Demographics only	• Physiological Vitals	• Clinical History	• e-Prescriptions
• Token Queue	• Triage Acuity	• Diagnoses	• FEFO Allocation
✗ No Clinical Notes	✗ No Prescribing	• Lab Orders & Meds	✗ No Consultation
✗ No Prescriptions	✗ No Consultations	• Clinical Notes	✗ No Lab Ordering

Key Principles of Platform Security:

1. Need-to-Know Privacy Boundaries:

Front-desk staff require citizen names and phone numbers to check patients in, but they have no legitimate clinical need to know that a citizen is undergoing treatment for depression or tuberculosis. The system strictly hides clinical notes from front desk screens.

2. Clinical Safety Boundaries:

Staff Nurses play a vital triage role, but they cannot issue electronic drug prescriptions. Doctors diagnose and prescribe, but they do not dispense drugs directly from the pharmacy. Each professional operates strictly within their regulatory and clinical scope.

3. Double-Entry Inventory Accountability:

Medicines cannot be added to or removed from clinic inventory informally. Stock enters via verified Goods Receipt Notes from approved suppliers and exits only through authenticated electronic doctor prescriptions.

4. Facility Isolation & Scoping:

Clinic staff can only access records, queues, and inventory belonging to their assigned facility. They cannot view patient queues or drug stocks from other clinics unless authorized by the District Health Officer for district-wide oversight.

5. Role-Based Audit Trail:

Every action in the system is stamped with the user's verified identity and their active operational role. If a nurse with dual front-desk and nursing roles registers a patient, it is audited under front-desk duty; when they enter vital signs, it is audited under nursing duty.

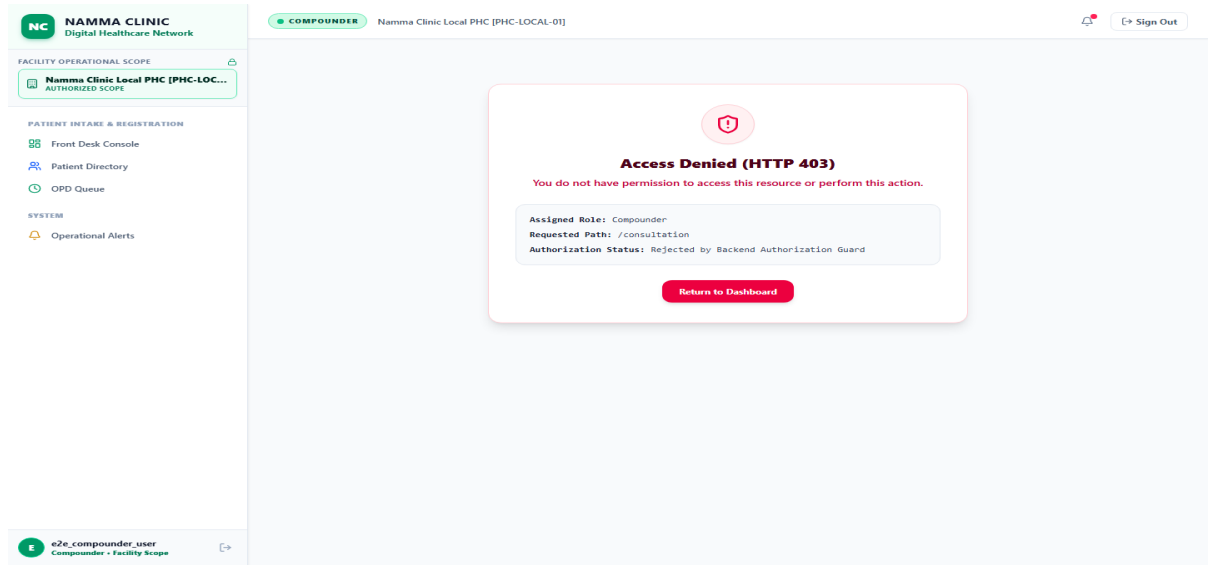


Figure 54 - Clean Role Separation Enforcement - Clear access denial dialog ensuring administrative staff cannot view confidential clinical consultation records or diagnostic history.

PART H - A REAL PATIENT JOURNEY: THE STORY OF MEENAKSHI AMMA

To illustrate how Namma Clinic works in daily healthcare operations, consider the visit of **Meenakshi Amma**, a 52-year-old resident who visits the clinic complaining of persistent headaches, fatigue, and blurred vision over the past two weeks.

Step 1: Arrival & Registration at Front Desk

Meenakshi walks into the clinic reception. Front Desk Officer **Kavitha** greets her and searches Meenakshi's mobile number in the patient directory. Finding no existing record, Kavitha opens the registration dialog, enters Meenakshi's demographic details, and issues outpatient **Token #T-104**. Kavitha gives Meenakshi her token slip and directs her to the triage waiting area.

- **Screen Used:** Citizen Intake & Token Registration.
- **Information Recorded:** Name, 52 yrs, Female, Phone number, Address.
- **What Happens Next:** Token T-104 immediately appears on Nurse Anitha's triage waiting queue.

Figure 55 - Step 1: Front Desk Registers Meenakshi - Reception intake dialog recording Meenakshi's demographic details and issuing sequential OPD Token T-104.

Step 2: Vital Signs Triage by Staff Nurse

Staff Nurse **Anitha** calls Token T-104 into the triage room. Anitha checks Meenakshi's blood pressure, pulse, temperature, oxygen saturation, height, and weight. The blood pressure reads **148/92 mmHg**, and a rapid fingerstick blood sugar reads **178 mg/dL**. The system automatically computes her BMI as **27.8** (Overweight) and assigns an **Amber (Urgent)** priority flag due to stage-1 hypertension.

- **Screen Used:** Nurse Triage & Vitals Assessment.
- **Information Recorded:** BP 148/92, Pulse 82 bpm, Temp 98.4°F, SpO2 98%, Random Blood Sugar 178 mg/dL, BMI 27.8, Acuity: Amber.
- **What Happens Next:** Meenakshi's token moves to Dr. Suresh's consultation queue, highlighted in amber with vitals pre-populated.

Figure 56 - Step 2: Nurse Anitha Measures Meenakshi's Vitals - Nursing triage assessment recording BP (148/92 mmHg), blood glucose (178 mg/dL), and BMI (27.8), triggering amber priority.

Step 3: Doctor Consultation, Examination & Prescribing

Medical Officer **Dr. Suresh** calls Meenakshi into the consultation room. Reviewing the elevated vitals recorded by Nurse Anitha, Dr. Suresh takes a detailed history, conducts an eye and cardiovascular examination, and confirms a clinical diagnosis of **Essential Hypertension (Primary)** and suspected **Type 2 Diabetes Mellitus**.

Dr. Suresh orders a rapid **Urine Albumin test** at the clinic lab and electronically prescribes **Amlodipine 5mg** (once daily in the morning for 30 days) and **Metformin 500mg** (once daily with food for 30 days). He schedules a follow-up revisit in 30 days.

- **Screen Used:** Medical Officer Consultation Workstation.
- **Information Recorded:** Symptoms (headache, blurred vision), Physical Exam findings, ICD Diagnoses, Urine Albumin lab requisition, electronic prescriptions for Amlodipine and Metformin, 30-day revisit order.
- **What Happens Next:** The urine test order appears instantly in the laboratory; the e-prescriptions appear on the pharmacy workstation.

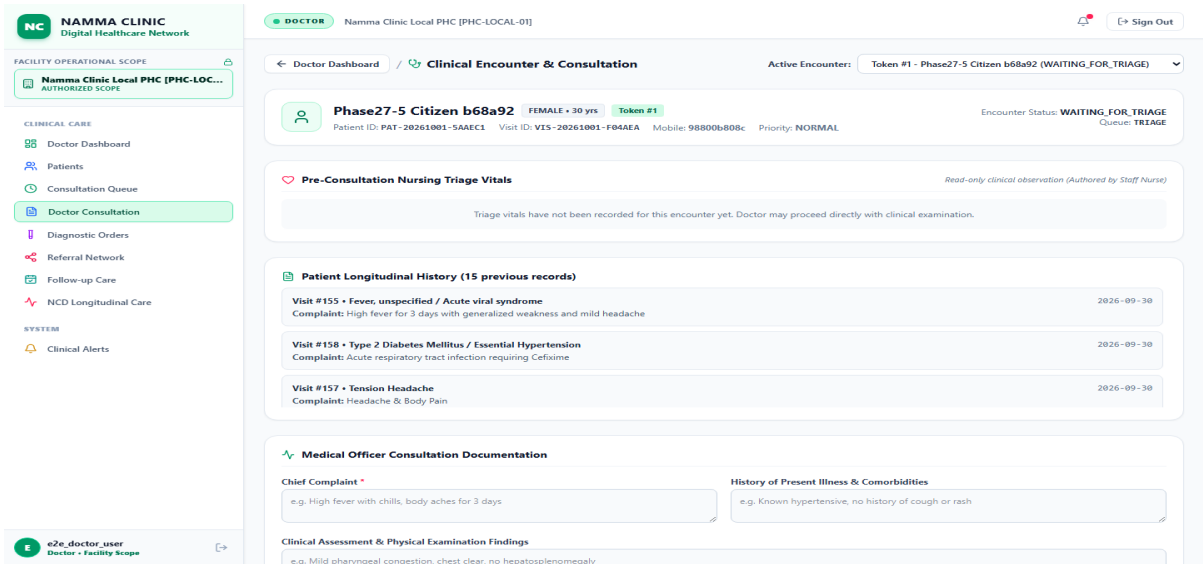


Figure 57 - Step 3: Dr. Suresh Records Meenakshi's Consultation - Clinician workspace documenting hypertension, ordering rapid urine albumin screening, and generating electronic prescriptions.

Step 4: Laboratory Sample Collection & Rapid Testing

Meenakshi walks to the diagnostic lab window. Diagnostic Lab Technician **Prakash** collects her urine sample, runs a rapid dipstick test, and enters the result (**Trace Albumin**) directly into the laboratory console.

- **Screen Used:** Diagnostic Laboratory Workstation.
- **Information Recorded:** Specimen collected, Test executed: Urine Albumin, Result: Trace, Status: Verified.
- **What Happens Next:** The result updates Dr. Suresh's chart immediately. With trace albumin noted, Dr. Suresh verifies that the prescribed Amlodipine remains clinically optimal and gives final electronic authorization.

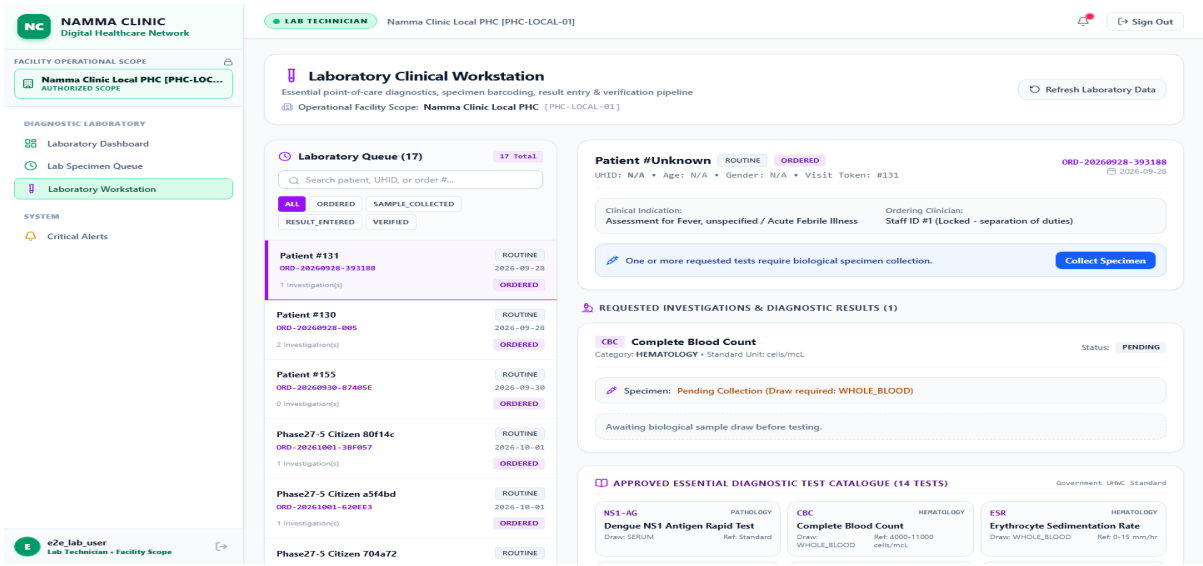


Figure 58 - Step 4: Lab Technician Prakash Enters Meenakshi's Test Results - Laboratory queue screen recording urine test results and immediately updating Dr. Suresh's consultation chart.

Step 5: Pharmacy Verification & FEFO Dispensing

Meenakshi arrives at the clinic dispensary window. Pharmacist **Deepa** opens Meenakshi's prescription on the pharmacy workstation. The system automatically selects active batches of Amlodipine 5mg and Metformin 500mg with the earliest expiration dates. Deepa scans the shelf, matches the batch numbers, prints label instructions in Tamil and English, and hands the medications to Meenakshi with dietary and dosage counseling.

- **Screen Used:** Pharmacy Dispensing Workstation.
- **Information Recorded:** Dispensed Batches (AML-2026-B1, MET-2026-A4), Expiry dates, Quantities (30 tablets each), counseling confirmed.
- **What Happens Next:** The clinic warehouse ledger automatically decrements 30 tablets of each medication. The prescription status changes to 'Dispensed'.

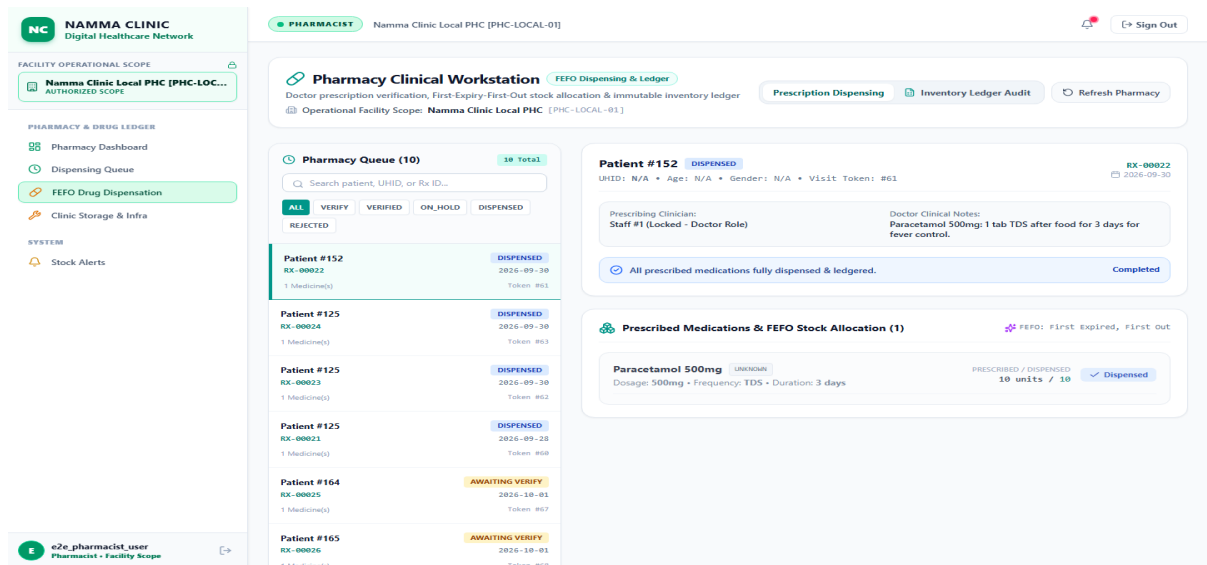


Figure 59 - Step 5: Pharmacist Deepa Verifies & Dispenses Medications - Pharmacy console confirming FEFO batch allocations, deducting stock from clinic inventory, and providing dosage counseling.

Step 6: Follow-up Scheduling & Departure

Before leaving, Meenakshi is entered into Nurse Anitha's chronic care follow-up calendar for review in four weeks. Her total visit time - from arrival at reception to walking out with her medications - was just 38 minutes.

Meenakshi returns home with clearly labeled medicines, a structured care plan, and an automated recall date, while clinic management gains complete visibility into every clinical and operational milestone of her visit.

PART I - CLIENT & STAKEHOLDER VALUE

Namma Clinic delivers concrete operational and clinical benefits across all stakeholder tiers:

STAKEHOLDER VALUE MATRIX		
CITIZENS & PATIENTS	CLINICAL TEAMS	HEALTH LEADERSHIP
• Fair queue numbers	• Legible e-prescriptions	• Live district
• Vitals safety checks	• Automatic BMI scoring	dashboards
• Zero expired drugs	• Immediate lab reports	• Outbreak alert
• Complete health record	• Reduced paperwork load	• Zero drug leak

Detailed Value Breakdown by Role

1. For Citizens & Urban Patients:
- **Fair, Dignified Care:** Automated queue tokens prevent queue jumping and crowding.
 - **Medication Quality:** Automated First-Expired, First-Out (FEFO) dispensing ensures citizens receive fresh, potent medicines.
 - **Continuity of Care:** Longitudinal records mean patients do not have to repeat their medical history every time they visit.

2. For Medical Officers (Doctors):

- **Faster Consultations:** Pre-recorded nursing vitals and BMI save clinician time.
- **Fewer Diagnostic Delays:** Laboratory orders and test results flow electronically.
- **Safer Prescribing:** Standardized drug lists eliminate dosing ambiguities and transcription errors.

3. For Staff Nurses:

- **Empowered Clinical Role:** Standardized vitals and acuity scoring allow nurses to actively prioritize urgent patients.
- **Chronic Disease Tracking:** NCD and wellness registers make follow-up care organized and manageable.
- **Elimination of Hand-Written Ledgers:** Replaces multiple paper logbooks with clean digital forms.

4. For Front Desk Officers:

- **Fast Citizen Intake:** Mobile number search brings up existing records in seconds.
- **Orderly Reception Area:** Sequential token issuance eliminates arguments at the reception window.
- **Protection from Clinical Disputes:** Restricted demographic screens keep front desk work purely clerical.

5. For Diagnostic Lab Technicians:

- **Clear Test Requests:** Legible electronic test orders eliminate confusion over hand-written test slips.
- **Integrated Quality Checks:** Automatic reference range highlighting flags abnormal samples immediately.
- **Audit-Ready Registers:** Test logs are stored electronically, ready for departmental inspection.

6. For Pharmacists:

- **Zero Dispensing Ambiguity:** Typed, structured prescriptions eliminate handwriting misinterpretation.
- **Automated Batch Selection:** System points directly to the earliest-expiring batch on shelf.
- **Accountability:** Real-time stock deductions protect the pharmacist from inventory discrepancies.

7. For Inventory Officers:

- **Procurement Control:** Structured purchase orders make vendor tracking straightforward.
- **Accurate Deliveries:** Goods Receipt Notes verify batch numbers, expiry dates, and delivered counts.
- **Stock Reconciliation:** Double-entry ledger prevents unexplained medicine losses.

8. For Clinic / Hospital Administrators:

- **Workforce Management:** Single-click role assignments ensure only qualified staff access each module.
- **Operational Insights:** Real-time queue dashboards identify clinic bottlenecks instantly.
- **Automated Reporting:** Daily outpatient and morbidity summaries generate automatically.

9. For District Health Officers (DHO):

- **Regional Visibility:** Live dashboard shows performance across all urban health centers.
- **Early Outbreak Warnings:** Syndromic surveillance highlights disease clusters days ahead of traditional reporting.
- **Optimized Resource Allocation:** Pinpoints clinics facing physician shortages or drug depletion for immediate support.

10. For Health Department & Government Leadership:

- **Demonstrated Return on Investment:** Higher patient throughput, lower administrative overhead, and minimized drug expiry waste.
- **Policy-Ready Health Data:** Comprehensive, standardized morbidity data for public health planning.
- **Local Infrastructure Resilience:** System runs reliably on local clinic hardware without expensive continuous cloud dependencies.

PART J - SYSTEM REFERENCE & GLOSSARY

1. Current Implemented Capabilities (Verified Operational Scope)

The table below confirms all functional capabilities currently built, operational, and verified on local clinic hardware:

Functional Area	Current Status	Operational Description
---	---	---
User Sign-In & Role Protection	Fully Operational	8 canonical operational roles; automatic boundary enforcement and security cards.
Front Desk Intake & Patient Registry	Fully Operational	Full citizen onboarding, deduplication, and daily sequential OPD token generation.

Nurse Triage & Clinical Vitals	Fully Operational	Physiological measurement (BP, pulse, temp, SpO2, RR, BMI) and 5-tier acuity scoring.
Doctor Consultation & EMR	Fully Operational	Clinical notes, symptom documentation, diagnosis entry, e-prescriptions, and lab orders.
Diagnostic Laboratory Workstation	Fully Operational	Specimen collection logging, numeric test value input, and report publishing.
Pharmacy Dispensing & FEFO	Fully Operational	Prescription verification, on-hold reasoning, and earliest-expiry batch stock deduction.
Warehouse Inventory & Procurement	Fully Operational	Supplier purchase orders, Goods Receipt Notes, batch tracking, and movement ledger.
Staff Lifecycle Administration	Fully Operational	Staff directory, role allocation, dual-role configuration, transfers, and suspensions.
Dual-Role Small Clinic Operations	Fully Operational	Nurse + Front Desk Officer and Pharmacist + Inventory multi-role assignment.
Executive Dashboards & Analytics	Fully Operational	District governance console, syndromic surveillance, and facility reporting.
Local Clinic Execution	Standard Operational Mode	Runs locally on standard clinic laptops and desktop workstations.

2. Planned Future Capabilities (Roadmap)

To ensure full transparency with stakeholders, the following features are strategically planned for future phases:

- **Planned Phase 29 - Live National Health Gateway Integration:**
 - *Status: Planned / Data Models Prepared.*
 - *Capability:* Connecting the current citizen data fields to live national digital health servers for automated OTP-based health ID generation and QR code check-in.
- **Planned Phase 30 - Telemedicine & Specialist Consultation:**
 - *Status: Planned.*
 - *Capability:* Integrated video consultation connecting urban clinic medical officers with hospital specialty consultants.
- **Planned Phase 31 - Centralized District Data Sync Gateway:**
 - *Status: Planned.*
 - *Capability:* Overnight encrypted batch synchronization transmitting local clinic databases to central state health department servers.
- **Planned Phase 32 - Barcode & QR Code Dispensing Hardware:**
 - *Status: Planned.*
 - *Capability:* Handheld barcode scanner integration for rapid medicine packaging verification at the pharmacy counter.

3. Glossary of Healthcare & Administrative Terms

- **Acuity Level:** A standardized assessment of how urgently a patient requires medical care (IMMEDIATE, VERY_URGENT, URGENT, STANDARD, NON_URGENT).
- **ARS (Arogya Raksha Samiti):** Untied health facility development and maintenance funds provided to public clinics for upkeep, emergency repairs, and citizen amenities.
- **Batch Number:** A unique identifier assigned by pharmaceutical manufacturers to a specific production lot of medication, essential for quality tracking and recall management.
- **BMI (Body Mass Index):** A calculated ratio of body weight to height (kg/m^2) used by nurses to screen for underweight, healthy, overweight, or obese status.
- **Chief Complaint:** The primary symptom or health concern expressed by the citizen upon arrival at the clinic.
- **DHO (District Health Officer):** The senior medical officer responsible for the administration and public health governance of all health centers across an administrative district.
- **Dual Role:** An administrative configuration allowing a single staff member in a small clinic to carry out two separate operational duties (e.g., Nurse + Front Desk Officer) without creating informal shortcuts.

- **Electronic Prescription (E-Rx):** A digitally created medication order specifying drug name, strength, dosage form, frequency, and duration.
- **EMR (Electronic Medical Record):** The longitudinal digital health file documenting a citizen's consultations, vitals, diagnoses, laboratory investigations, and medication history.
- **FEFO (First-Expired, First-Out):** A strict inventory management discipline requiring that medication batches expiring earliest are dispensed before newer stock, preventing drug wastage.
- **Front Desk Officer:** The administrative reception worker responsible for citizen registration, demographic search, and daily OPD token issuance.
- **GRN (Goods Receipt Note):** A certified administrative document verifying that delivered pharmaceuticals have been inspected, checked against purchase orders, and added to clinic stock.
- **NCD (Non-Communicable Diseases):** Long-term chronic medical conditions (e.g., Hypertension, Type-2 Diabetes) requiring periodic monitoring and regular medication refills.
- **OPD (Outpatient Department):** Ambulatory care services provided to walk-in citizens who receive diagnosis and treatment without requiring overnight hospital admission.
- **PHC (Primary Health Center):** A frontline government healthcare facility providing comprehensive primary medical care, maternal health, immunization, and disease prevention.
- **Purchase Order (PO):** A formal procurement document sent to an authorized pharmaceutical supplier requesting specific medicines, quantities, and agreed delivery terms.
- **SpO2 (Blood Oxygen Saturation):** The percentage of oxygen-saturated hemoglobin in the blood measured non-invasively using a pulse oximeter.
- **Syndromic Surveillance:** The continuous collection and analysis of presenting symptoms (e.g., acute fever, rash, respiratory distress) to detect communicable disease outbreaks early.
- **Triage:** The standardized process of examining walk-in patients immediately upon arrival to determine medical urgency and prioritize clinician examination.
- **Vitals:** Core physiological indicators of health status measured by the nurse: Blood Pressure, Heart Rate, Body Temperature, Respiratory Rate, and Oxygen Saturation.
- **Void Token:** The administrative cancellation of an OPD token issued in error, permitted only before clinical triage has begun.